

HANDS-ON LAB GUIDE

Learning Windows Server 2025 and Active Directory

From first install to a hardened enterprise directory. Every exercise is performed in an isolated VirtualBox lab for the fictional company Greenfield Manufacturing.

DOMAIN
greenfieldcorp.lab

NETBIOS
GREENFIELD

PLATFORM
Oracle VirtualBox 7.x

Contents

FRONT MATTER • LABS 01–32 • COMMAND CHEAT SHEET

A Using this guide

B Lab topology & IP plan

C Naming conventions

D Snapshot map

PART I – FOUNDATIONS AND THE LAB

01 Survey Your Platform and Choose a Lab Host

02 Build the Lab Environment

03 Install Windows Server 2025

04 Server Fundamentals and Initial Configuration

05 PowerShell for Server Administrators

PART II – ACTIVE DIRECTORY CORE

06 What Active Directory Actually Is

07 Design the Logical Structure

08 Sites, Replication, and the Physical Structure

09 Promote the First Domain Controller

10 DNS, the Foundation Underneath Everything

11 FSMO Roles and the Global Catalog

PART III – IDENTITY, POLICY, AND ACCESS

12 Users, Groups, and Computers

13 Group Policy Fundamentals

14 Group Policy in Depth

15 Service Accounts and Managed Identities

16 Delegation of Administration

17 Authentication Protocols: Kerberos and NTLM

PART IV – SECURITY AND HARDENING

18 Securing the Directory

19	Understanding Active Directory Attacks
20	Detecting and Defending Against AD Attacks
21	Certificate Services and PKI Basics
22	Windows LAPS and Local Account Security
PART V – INFRASTRUCTURE SERVICES AND LIFECYCLE	
23	DHCP and IP Address Management
24	File Services, Storage, and Permissions
25	Hyper-V and Virtualization
26	Backup, Recovery, and Business Continuity
27	Upgrading and Migrating to Windows Server 2025
PART VI – AUTOMATION, MONITORING, AND MASTERY	
28	Automating Administration with PowerShell
29	Monitoring, Health, and Performance
30	Hybrid Identity with Entra ID
31	Troubleshooting Methodology
32	Design, Best Practices, and Career Direction
E	Command cheat sheet

A Using this guide

Each lab maps to one video lesson and builds on the state left by the previous lab. Work in order. Every lab uses the same structure:

OBJECTIVES	What you will be able to do when finished.
PREREQUISITES	Machines that must be running and the snapshot to start from.
TASKS	Numbered steps with exact commands. The machine to run on is shown in brackets, e.g. [DC01].
EXPECTED OUTPUT	What success looks like. Values such as GUIDs and dates will differ.
TROUBLESHOOTING	Common failures and fixes.
KNOWLEDGE CHECK	Questions to answer before moving on.

Conventions. Unless stated otherwise, run commands in an elevated PowerShell 5.1 session. PS> prompts are omitted. Lab password for all accounts: P@ssw0rd.Lab25! — never reuse it outside the lab.

WARNING – ISOLATION

Nothing in this guide touches production. Attack-simulation labs (19, 20) must only be performed inside the isolated lab network. Never connect lab DCs to a bridged adapter.

B Lab topology & IP plan

All VMs attach to one VirtualBox **NAT Network** named GFM-NAT. It provides outbound internet through VirtualBox's built-in router at 10.10.10.1 while blocking inbound traffic from your home or office LAN. VirtualBox DHCP is disabled on this network so that DC01/SRV01 can serve DHCP in Lab 23.

Setting	Value
NAT Network	GFM-NAT · 10.10.10.0/24 · DHCP off
Default gateway	10.10.10.1 (VirtualBox NAT router)
DNS (all domain members)	10.10.10.10, 10.10.10.11
DNS forwarder (on DCs)	1.1.1.1, 9.9.9.9
Branch site subnet (logical)	10.10.20.0/24 · site BRANCH-DET
DHCP scope (Lab 23)	10.10.10.100 – 10.10.10.199

Host	IPv4	OS / install	vCPU/RAM/Disk	Role	Built
DC01	10.10.10.10	WS2025 Std, Desktop	2/4 GB/60 GB	First DC, DNS, PDC Emulator, GC	L03
SRV02	10.10.10.21	WS2025 Std, Core	2/2 GB/40 GB	Member: gMSA host, DHCP partner	L03
DC02	10.10.10.11	WS2025 Std, Desktop	2/3 GB/60 GB	Second DC, DNS, GC	L11
SRV01	10.10.10.20	WS2025 Std, Desktop	2/4 GB/80 GB	Member: files, DHCP, issuing CA, Hyper-V	L12
CL01	10.10.10.50 → DHCP	Windows 11 Ent eval	2/4 GB/64 GB	Domain client	L12
ROOTCA	10.10.10.30	WS2025 Std, Core, workgroup	1/2 GB/40 GB	Offline root CA	L21
DC03	10.10.10.12	WS2025 Std, Desktop	2/3 GB/60 GB	Migration target DC	L27

Host sizing: 16 GB RAM runs DC01 + SRV02 + CL01 comfortably; 32 GB is recommended from Lab 11 onward. Power off machines a lab does not list.

C Naming conventions

Object	Pattern	Example
Domain controller	DC###	DC01
Member server	SRV###	SRV01
Workstation	CL###	CL01
Standard user	first.last	maria.lopez
Admin account	adm-t#-first.last	adm-t0-maria.lopez
Role (global) group	GG-<Dept>-<Role>	GG-ENG-Designers
Resource (domain local) group	DL-<Resource>-<Access>	DL-ENG-Share-Modify
Universal group	UG-<Purpose>	UG-AllEngineering
GPO	<Scope>-<C U>-<Purpose>-v#	WKS-U-DesktopLockdown-v1
gMSA / dMSA	gmsa-<app> / dmsa-<app>	gmsa-webapp
Snapshot	L###-<START END>-<state>	L09-END-dc01-promoted

OU hierarchy (built in Lab 07)

```
greenfieldcorp.lab
├── OU=Greenfield
│   ├── OU=Admin           ── OU=Tier0 · OU=Tier1 · OU=Tier2
│   ├── OU=Users           ── OU=Engineering · OU=Production · OU=Finance ·
├── OU=Sales · OU=IT
│   ├── OU=Groups          ── OU=Role · OU=Resource
│   ├── OU=Computers       ── OU=Workstations · OU=Servers
│   ├── OU=ServiceAccounts
│   └── OU=Disabled
```

D

Snapshot map

Take the END snapshot on every listed VM after each lab. To retry a lab, restore its "Start from" snapshot. Delete snapshots older than three labs to save disk. * Lab 08 is performed after Lab 09.

Lab	Start from	Take at end	VMs
01	–	– (no snapshot)	host only
02	–	L02-END-template-sysprepped	TPL-WS2025
03	L02-END-template-sysprepped	L03-END-fresh-install	DC01, SRV02
04	L03-END-fresh-install	L04-END-net-configured	DC01, SRV02
05	L04-END-net-configured	L05-END-ps-report	DC01, SRV02
06	L05-END-ps-report	– (no snapshot)	none
07	L05-END-ps-report	– (no snapshot)	none
08 *	L09-END-dc01-promoted	L08-END-sites	DC01
09	L05-END-ps-report	L09-END-dc01-promoted	DC01
10	L08-END-sites	L10-END-dns	DC01, SRV02
11	L10-END-dns	L11-END-dc02-fsmo	DC01, DC02
12	L11-END-dc02-fsmo	L12-END-users-groups	DC01, DC02, SRV01, SRV02, CL01
13	L12-END-users-groups	L13-END-gpo-basics	DC01, CL01
14	L13-END-gpo-basics	L14-END-gpo-depth	DC01, SRV01, CL01
15	L14-END-gpo-depth	L15-END-managed-accounts	DC01, SRV02
16	L15-END-managed-accounts	L16-END-delegation	DC01, CL01
17	L16-END-delegation	L17-END-kerberos	DC01, SRV01, SRV02, CL01
18	L17-END-kerberos	L18-END-baseline	DC01, DC02, CL01
19	L18-END-baseline	L19-END-attack-observed	DC01, CL01
20	L19-END-attack-observed	L20-END-detection	DC01, DC02
21	L20-END-detection	L21-END-pki	ROOTCA, SRV01, DC01
22	L21-END-pki	L22-END-laps	DC01, SRV01, SRV02
23	L22-END-laps	L23-END-dhcp	DC01, SRV01, SRV02, CL01
24	L23-END-dhcp	L24-END-file-services	DC01, SRV01, CL01
25	L24-END-file-services	L25-END-hyperv	SRV01, SRV02, DC01
26	L25-END-hyperv	L26-END-recovery	DC01, DC02
27	L26-END-recovery	L27-END-migrated-32k	DC01, DC02, DC03
28	L27-END-migrated-32k	L28-END-automation	DC01, DC03, SRV01
29	L28-END-automation	L29-END-monitoring	DC01, DC03, SRV01
30	L29-END-monitoring	L30-END-hybrid	DC01, DC03, SRV01
31	L30-END-hybrid	L31-END-troubleshooting	DC01, DC03, SRV01, CL01
32	L31-END-troubleshooting	L32-END-course-complete	all

Foundations and the Lab

Survey Your Platform and Choose a Lab Host

Survey your hardware or cloud options and choose a lab platform.

OBJECTIVES

- Inventory host CPU, RAM, disk, and virtualization support
- Detect conflicts between VirtualBox and Windows hypervisor features
- Record a lab platform decision and capacity budget

PREREQUISITES

A Windows 10/11 host (macOS/Linux hosts: use the equivalent system tools).
Local administrator rights. No VMs yet.

TASKS

- 1 **Open an elevated PowerShell on the host.** Right-click Start > Terminal (Admin).
- 2 **Inventory CPU and virtualization support.** VirtualizationFirmwareEnabled must be True. If it is False, enable Intel VT-x or AMD-V/SVM in UEFI setup.

```
Get-CimInstance Win32_Processor |  
    Select-Object Name, NumberOfCores, NumberOfLogicalProcessors,  
    VirtualizationFirmwareEnabled, SecondLevelAddressTranslationExtensions
```

- 3 **Inventory memory and free disk.** Plan for 16 GB RAM minimum and 250 GB free SSD space.

```
[math]::Round((Get-CimInstance Win32_ComputerSystem).TotalPhysicalMemory/1GB,1)  
Get-Volume | Where-Object DriveLetter | Select-Object DriveLetter,  
FileSystemLabel,  
    @{n='FreeGB';e={[math]::Round($_.SizeRemaining/1GB)}} , @{n='SizeGB';e=  
    {[math]::Round($_.Size/1GB)}}}
```

- 4 **Check for Windows hypervisor features.** When Hyper-V, Virtual Machine Platform, or Memory Integrity is on, VirtualBox runs VMs through the Windows hypervisor, which is slower and can block nested virtualization (Lab 25). Note what is on; turning these off is optional.

```
Get-WindowsOptionalFeature -Online |
  Where-Object FeatureName -in 'Microsoft-Hyper-V-
All','VirtualMachinePlatform','HypervisorPlatform' |
  Select-Object FeatureName, State
bcdedit /enum '{current}' | Select-String hypervisorlaunchtype
Get-CimInstance -Namespace root\Microsoft\Windows\DeviceGuard -ClassName
Win32_DeviceGuard |
  Select-Object VirtualizationBasedSecurityStatus
```

- 5 **Check the host TPM (informational).** The lab VMs get a virtual TPM 2.0 from VirtualBox 7, so this only matters for Secured-core discussions.

```
Get-Tpm | Select-Object TpmPresent, TpmReady, ManufacturerVersion
```

- 6 **Install VirtualBox 7.1 or later.** Download VirtualBox and the matching Extension Pack from [virtualbox.org](https://www.virtualbox.org). Install both, then confirm the version.

```
& "$env:ProgramFiles\Oracle\VirtualBox\VBoxManage.exe" --version
[Environment]::SetEnvironmentVariable('Path', $env:Path +
";$env:ProgramFiles\Oracle\VirtualBox", 'User')
```

- 7 **Complete the platform worksheet.** Record: host RAM, free disk, VT-x state, hypervisor conflicts, chosen platform (VirtualBox on host, or a cloud alternative), and the VMs you can run concurrently using the Topology table in section B.

EXPECTED OUTPUT

```
Name : Intel(R) Core(TM) i7-1260P
NumberOfLogicalProcessors : 16
VirtualizationFirmwareEnabled : True
32.0
VBoxManage --version -> 7.1.x r16xxxx
```

TROUBLESHOOTING

VirtualizationFirmwareEnabled is False	Enter UEFI setup and enable Intel VT-x / AMD SVM. On some laptops it is labeled "Virtualization Technology".
VirtualBox shows a green turtle icon on running VMs	The Windows hypervisor is active. Either accept the slower performance or disable Hyper-V/VMP and run <code>bcdedit /set hypervisorlaunchtype off</code> , then reboot.
Less than 16 GB RAM on host	Run DC01 and one other VM at a time, or use a cloud lab (Azure/AWS) and follow the same IP plan.

KNOWLEDGE CHECK

- Q1 What workload differences separate a server OS from a desktop OS?
- Q2 Which edition grants unlimited Windows Server guest VMs, and why does that matter for virtualization hosts?
- Q3 When does Windows Server 2025 mainstream support end, and why should that date shape your planning?

SNAPSHOT ▶ none – no VM changes in this lab

LAB 02

LESSON 2

Build the Lab Environment

Stand up the hypervisor, create the virtual network, and build a base template.

OBJECTIVES

- Create an isolated VirtualBox NAT Network for the lab
- Download and verify the Windows Server 2025 evaluation ISO
- Build and sysprep a reusable base template VM
- Apply the snapshot naming convention

PREREQUISITES

VirtualBox 7.1+ and Extension Pack installed (Lab 01). 60 GB free disk.

TASKS

- 1 **Create the lab network.** A NAT Network gives outbound internet but no inbound path from your LAN. DHCP is disabled because the domain will provide it later.

```
VBoxManage natnetwork add --netname GFM-NAT --network "10.10.10.0/24" --enable --  
dhcp off --ipv6 off  
VBoxManage natnetwork list
```

- 2 **Download the evaluation ISO.** From the Microsoft Evaluation Center, download Windows Server 2025 (ISO, English). Save it as C:\ISO\WS2025_EVAL.iso.

- 3 **Verify the ISO hash.** Compare to the SHA256 published on the download page. Do not use an ISO that does not match.

```
Get-FileHash C:\ISO\WS2025_EVAL.iso -Algorithm SHA256
```

- 4 **Create the template VM.** UEFI firmware, TPM 2.0, and Secure Boot are required for the Secured-core defaults covered in Lesson 1.

```
$vm='TPL-WS2025'
VBoxManage createvm --name $vm --ostype Windows2025_64 --register
VBoxManage modifyvm $vm --cpus 2 --memory 4096 --vram 128 --firmware efi --tpm-
type 2.0 --graphicscontroller vboxsvga
VBoxManage modifyvm $vm --nic1 natnetwork --nat-network1 GFM-NAT --nictype1
82540EM
VBoxManage createmedium disk --filename "$HOME\VirtualBox VMS\$vm\$vm.vdi" --size
61440
VBoxManage storagectl $vm --name SATA --add sata --controller IntelAhci
VBoxManage storageattach $vm --storagectl SATA --port 0 --type hdd --medium
"$HOME\VirtualBox VMS\$vm\$vm.vdi"
VBoxManage storageattach $vm --storagectl SATA --port 1 --type dvddrive --medium
C:\ISO\WS2025_EVAL.iso
VBoxManage modifynvram $vm inituefivarstore
VBoxManage modifynvram $vm enrollmssignatures
VBoxManage modifynvram $vm enrollorclpk
```

- 5 **Install Windows Server 2025 Standard Evaluation (Desktop Experience).** Start the VM, press a key to boot from DVD, choose Custom install on the empty disk. Set the Administrator password to the lab password.

```
VBoxManage startvm TPL-WS2025
```

- 6 **Install Guest Additions and patch.** Devices > Insert Guest Additions CD image, run VBoxWindowsAdditions.exe, reboot. Then run Windows Update until no updates remain.

```
Start-Process ms-settings:windowsupdate
```

- 7 **Generalize with sysprep.** Inside the VM. The VM shuts down when finished; do NOT boot it again or it will specialize.

```
C:\Windows\System32\Sysprep\sysprep.exe /generalize /oobe /shutdown /mode:vm
```

- 8 **Snapshot the template and detach the ISO.** Clones will be linked to this snapshot to save disk.

```
VBoxManage storageattach TPL-WS2025 --storagectl SATA --port 1 --type dvddrive --
medium emptydrive
VBoxManage snapshot TPL-WS2025 take "L02-END-template-sysprepped" --description
"Generalized WS2025 DE, patched"
```

EXPECTED OUTPUT

```
NetworkName: GFM-NAT
Network:     10.10.10.0/24
Gateway:     10.10.10.1
DHCP Server: No
Snapshot taken. UUID: 4f1c...
```

TROUBLESHOOTING

VBoxManage: not recognized	Open a new terminal after adding the VirtualBox folder to Path, or call it by full path.
Setup says "This PC can't run Windows Server"	TPM or Secure Boot missing. Re-run the modifyvm --tpm-type and modifynvram commands with the VM powered off.
Sysprep fails: "unable to validate your Windows installation"	Check C:\Windows\System32\Sysprep\Panther\setupact.log. Commonly caused by a pending update; reboot, finish updates, retry.

KNOWLEDGE CHECK

Q1 Why is a NAT Network safer for an AD lab than a bridged adapter?

Q2 What does /generalize remove, and why must every clone be generalized?

Q3 How many days does the evaluation run, and how can it be extended with slmgr /rearm?

SNAPSHOT ▶ L02-END-template-sysprepped (TPL-WS2025)

Install Windows Server 2025

Install one Desktop Experience server and one Server Core server from your template.

OBJECTIVES	• Clone a Desktop Experience server from the template • Perform a clean Server Core installation from ISO • Set time zone and verify licensing state • Capture a first-boot health baseline and use sconfig
PREREQUISITES	Snapshot L02-END-template-sysprepped on TPL-WS2025. Verified ISO at C:\ISO\WS2025_EVAL.iso.

TASKS

- 1
- Create the Desktop Experience server as a linked clone. This VM becomes DC01. Complete OOBE with the lab password.

```
VBoxManage clonevm TPL-WS2025 --snapshot "L02-END-template-sysprepped" --options
link --name DC01 --register
VBoxManage modifyvm DC01 --memory 4096
VBoxManage startvm DC01
```

- 2 **Create the Server Core VM from ISO.** Choose "Windows Server 2025 Standard Evaluation" (no "Desktop Experience" in the name).

```
$vm='SRV02'
VBoxManage createvm --name $vm --ostype Windows2025_64 --register
VBoxManage modifyvm $vm --cpus 2 --memory 2048 --firmware efi --tpm-type 2.0 --
nic1 natnetwork --nat-network1 GFM-NAT
VBoxManage createmedium disk --filename "$HOME\VirtualBox VMS\$vm\$vm.vdi" --size
40960
VBoxManage storagectl $vm --name SATA --add sata
VBoxManage storageattach $vm --storagectl SATA --port 0 --type hdd --medium
"$HOME\VirtualBox VMS\$vm\$vm.vdi"
VBoxManage storageattach $vm --storagectl SATA --port 1 --type dvddrive --medium
C:\ISO\WS2025_EVAL.iso
VBoxManage modifynvram $vm inituefivarstore; VBoxManage modifynvram $vm
enrollmssignatures; VBoxManage modifynvram $vm enrollorclpk
VBoxManage startvm $vm
```

- 3 **Confirm the disk layout. [both]** UEFI installs create EFI, MSR, Windows, and Recovery partitions on a GPT disk.

```
Get-Disk | Select-Object Number, PartitionStyle, Size
Get-Partition | Select-Object PartitionNumber, Type, @{n='GB';e=
{[math]::Round($_.Size/1GB,1)}}}
```

- 4 **Set time zone and check time. [both]** Kerberos tolerates only 5 minutes of skew; correct time now avoids later failures.

```
Set-TimeZone -Id 'Eastern Standard Time'
w32tm /query /status
Get-Date
```

- 5 **Check licensing and evaluation clock. [both]** Note days remaining. To convert later: DISM /Online /Set-Edition:ServerStandard /ProductKey:<key> /AcceptEula.

```
slmgr /dlv
DISM /Online /Get-CurrentEdition
DISM /Online /Get-TargetEditions
```

- 6 **Review 2025 security defaults. [DC01]** Confirm VBS, Secure Boot, and SMB signing requirement are on by default.

```
Confirm-SecureBootUEFI
Get-SmbClientConfiguration | Select-Object RequireSecuritySignature
Get-SmbServerConfiguration | Select-Object RequireSecuritySignature, EncryptData
Get-CimInstance -Namespace root\Microsoft\Windows\DeviceGuard Win32_DeviceGuard |
Select-Object VirtualizationBasedSecurityStatus
```

- 7 **Capture a health baseline. [both]** Save the output to C:\Baseline for comparison after later labs.

```
New-Item C:\Baseline -ItemType Directory -Force | Out-Null
Get-ComputerInfo | Select-Object OsName, OsVersion, OsBuildNumber,
CsTotalPhysicalMemory, BiosFirmwareType |
    Out-File C:\Baseline\os.txt
Get-Service | Where-Object StartType -eq Automatic | Where-Object Status -ne
Running |
    Out-File C:\Baseline\stopped-auto-services.txt
Get-WinEvent -LogName System -MaxEvents 50 | Where-Object LevelDisplayName -in
'Error','Critical' |
    Out-File C:\Baseline\system-errors.txt
```

- 8 **Explore sconfig. [SRV02]** Server Core launches sconfig at logon. Note the menu numbers for network (8), updates (6), and remote management (4). Press 15 to exit to PowerShell; type sconfig to return.

```
sconfig
```

EXPECTED OUTPUT

```
PartitionStyle : GPT
Confirm-SecureBootUEFI -> True
RequireSecuritySignature : True    (SMB client and server)
OsName           : Microsoft Windows Server 2025 Standard Evaluation
BiosFirmwareType : Uefi
slmgr: License Status: Initial grace period / Timebased activation expiration: 259200
minutes
```

TROUBLESHOOTING

Clone boots to "Windows could not complete the installation"	The template was booted after sysprep. Restore L02-END and re-run sysprep.
Server Core has no mouse integration	Expected. Use keyboard, or install Guest Additions from D:\VBoxWindowsAdditions.exe /S.

Confirm-SecureBootUEFI returns False

NVRAM keys were not enrolled. Power off, run the modifynvram enroll commands, start again.

KNOWLEDGE CHECK

- Q1 List two tasks that are harder on Server Core and two advantages that justify it.
- Q2 Why does accurate time matter before a server ever joins a domain?
- Q3 What changed in SMB signing defaults compared with Windows Server 2022?

SNAPSHOT ▸ L03-END-fresh-install (DC01, SRV02)

LAB 04

LESSON 4

Server Fundamentals and Initial Configuration

Configure static networking, rename both servers, and enable remote management.

OBJECTIVES

- Assign static IPv4 addressing per the IP plan
- Rename servers to the naming standard
- Configure firewall rules and update behavior
- Enable and test WinRM and Remote Desktop

PREREQUISITES

DC01 and SRV02 from snapshot L03-END-fresh-install.

TASKS

1 Identify the adapter. [both]

```
Get-NetAdapter | Select-Object Name, InterfaceIndex, Status, MacAddress
```

2 Set static IP on DC01. [DC01] A future DC must have a static address and point DNS at itself (loopback after promotion). Before promotion, use its own IP.

```
New-NetIPAddress -InterfaceAlias 'Ethernet' -IPAddress 10.10.10.10 -PrefixLength 24 -DefaultGateway 10.10.10.1
Set-DnsClientServerAddress -InterfaceAlias 'Ethernet' -ServerAddresses 10.10.10.10,127.0.0.1
Rename-NetAdapter -Name 'Ethernet' -NewName 'LAN'
Disable-NetAdapterBinding -Name 'LAN' -ComponentID ms_tcpip6 -WhatIf # leave IPv6 enabled; review only
```

3 Set static IP on SRV02. [SRV02] Member servers point DNS at the DC(s), never at a public resolver.


```
New-NetIPAddress -InterfaceAlias 'Ethernet' -IPAddress 10.10.10.21 -PrefixLength 24 -DefaultGateway 10.10.10.1
Set-DnsClientServerAddress -InterfaceAlias 'Ethernet' -ServerAddresses 10.10.10.10
```

4 **Rename both servers. [each]**

```
Rename-Computer -NewName DC01 -Restart # on DC01
Rename-Computer -NewName SRV02 -Restart # on SRV02
```

5 **Review firewall profiles. [both]** Unidentified networks default to Public, which blocks most management traffic. Allow ICMP for lab diagnostics.

```
Get-NetConnectionProfile
Get-NetFirewallProfile | Select-Object Name, Enabled, DefaultInboundAction
Set-NetConnectionProfile -InterfaceAlias LAN -NetworkCategory Private # DC01
(Ethernet on SRV02)
Enable-NetFirewallRule -DisplayName 'File and Printer Sharing (Echo Request - ICMPv4-In)'
```

6 **Enable remote management. [both]** WinRM is on by default for Private/Domain. Enable RDP on DC01 with NLA.

```
Enable-PSRemoting -Force
Set-ItemProperty 'HKLM:\System\CurrentControlSet\Control\Terminal Server' -Name fDenyTSConnections -Value 0
Enable-NetFirewallRule -DisplayGroup 'Remote Desktop'
```

7 **Trust workgroup peers for WinRM (pre-domain only). [DC01]** Needed until both machines join the domain. Remove after Lab 12.

```
Set-Item WSMAN:\localhost\Client\TrustedHosts -Value 'SRV02,10.10.10.21' -Force
Test-WSMan 10.10.10.21
Invoke-Command -ComputerName 10.10.10.21 -Credential (Get-Credential SRV02\Administrator) -ScriptBlock { hostname; Get-NetIPAddress -AddressFamily IPv4 | Select IPAddress }
```

8 **Add SRV02 to Server Manager. [DC01]** Server Manager > Manage > Add Servers > DNS tab > 10.10.10.21. Right-click > Manage As to supply credentials.

9 **Review update settings and hotpatch eligibility. [DC01]** Hotpatch on 2025 Standard/Datacenter requires Azure Arc enrollment and VBS. Record the current state only.

```
Get-HotFix | Sort-Object InstalledOn -Descending | Select-Object -First 5
(Get-CimInstance -Namespace root\Microsoft\Windows\DeviceGuard Win32_DeviceGuard).VirtualizationBasedSecurityStatus
```

EXPECTED OUTPUT

```
IPAddress : 10.10.10.10 PrefixLength : 24
NetworkCategory : Private
wsmid      : http://schemas.dmtf.org/wbem/wsman/identity/1/wsmanidentity.xsd
SRV02
IPAddress: 10.10.10.21
```

TROUBLESHOOTING

New-NetIPAddress: "Instance already exists"	An APIPA/previous address exists. Remove-NetIPAddress -InterfaceAlias Ethernet -Confirm:\$false, then retry.
Test-WSMan fails with WinRM cannot complete the operation	Profile is Public or ping blocked. Set profile to Private on both machines and re-run Enable-PSRemoting.
No internet access	Confirm gateway 10.10.10.1 and that GFM-NAT is enabled. DNS 10.10.10.10 will not resolve externally until the forwarder in Lab 10 — temporarily add 1.1.1.1 to SRV02 only.

KNOWLEDGE CHECK

- Q1 Why must a domain controller never obtain its IP from DHCP?
- Q2 What is the difference between a role and a feature?
- Q3 Which firewall profile applies to an unidentified network, and what does that block?

SNAPSHOT ▸ L04-END-net-configured (DC01, SRV02)

LAB 05

LESSON 5

PowerShell for Server Administrators

Write a script that reports configuration and health across both lab servers.

OBJECTIVES

- Use Get-Help, Get-Command, and Get-Member to discover cmdlets
- Filter, select, and format pipeline objects
- Write a parameterized script with error handling
- Run it one-to-many with PowerShell remoting

PREREQUISITES

DC01 and SRV02 from L04-END-net-configured. TrustedHosts set on DC01.

TASKS

1 Update help and explore. [DC01]

```
Update-Help -ErrorAction SilentlyContinue
Get-Command -Verb Get -Noun *Service*
Get-Help Get-Service -Examples
Get-Service | Get-Member -MemberType Property
```

2 Practice the pipeline. Objects flow through the pipeline; format only at the very end.

```
Get-Service | Where-Object { $_.Status -eq 'Running' -and $_.StartType -eq
'Automatic' } |
    Sort-Object DisplayName | Select-Object -First 10 Name, DisplayName | Format-
Table -AutoSize
Get-Process | Sort-Object WS -Descending | Select-Object -First 5 Name,
@{n='WS_MB';e={[int]($_.WS/1MB)}}}
```

3 Practice variables, arrays, and hash tables.

```
$servers = @('localhost','10.10.10.21')
$info = @{ Owner='IT'; Site='HQ-GRR'; Tier=1 }
$info.Keys; $info['Site']
$servers.Count
```

4 Check execution policy. RemoteSigned is the server default. Do not set Unrestricted or Bypass machine-wide.

```
Get-ExecutionPolicy -List
New-Item C:\Scripts -ItemType Directory -Force
```

5 Create C:\Scripts\Get-LabHealth.ps1. [DC01] Save with notepad C:\Scripts\Get-LabHealth.ps1.

```

[CmdletBinding()]
param(
    [string[]]$ComputerName = @('localhost'),
    [pscredential]$Credential,
    [string]$OutFile = "C:\Scripts\LabHealth-$(Get-Date -Format yyyyMMdd-HH:mm).csv"
)
$block = {
    $os = Get-CimInstance Win32_OperatingSystem
    $disk = Get-CimInstance Win32_LogicalDisk -Filter "DeviceID='C:'"
    $ip = (Get-NetIPAddress -AddressFamily IPv4 | Where-Object IPAddress -notlike '127.*').IPAddress -join ','
    $dns = (Get-DnsClientServerAddress -AddressFamily IPv4 | Where-Object ServerAddresses).ServerAddresses -join ','
    $stopped = Get-Service | Where-Object { $_.StartType -eq 'Automatic' -and $_.Status -ne 'Running' }
    [pscustomobject]@{
        Computer      = $env:COMPUTERNAME
        OS             = $os.Caption
        Build          = $os.BuildNumber
        UptimeHours    = [int]((Get-Date) - $os.LastBootUpTime).TotalHours
        IPv4           = $ip
        DNS            = $dns
        FreeC_GB       = [math]::Round($disk.FreeSpace/1GB,1)
        FreeMemPct     = [int](100 * $os.FreePhysicalMemory /
$os.TotalVisibleMemorySize)
        StoppedAuto    = ($stopped.Name -join ',')
    }
}
$results = foreach ($c in $ComputerName) {
    try {
        $p = @{ ComputerName = $c; ScriptBlock = $block; ErrorAction = 'Stop' }
        if ($Credential -and $c -ne 'localhost') { $p.Credential = $Credential }
        Invoke-Command @p | Select-Object * -ExcludeProperty PSComputerName,
RunspaceId, PSShowComputerName
    } catch {
        Write-Warning "$c unreachable: $($_.Exception.Message)"
        [pscustomobject]@{ Computer = $c; OS = 'UNREACHABLE' }
    }
}
$results | Export-Csv $OutFile -NoTypeInformation
$results | Format-Table Computer, Build, UptimeHours, IPv4, FreeC_GB, FreeMemPct -
AutoSize
Write-Verbose "Saved to $OutFile"

```

6 Run it one-to-many. [DC01]

```
$cred = Get-Credential SRV02\Administrator
C:\Scripts\Get-LabHealth.ps1 -ComputerName localhost,10.10.10.21 -Credential $cred
-Verbose
```

- 7 **Test error handling.** Include a host that does not exist and confirm the script warns and continues.

```
C:\Scripts\Get-LabHealth.ps1 -ComputerName localhost,10.10.10.99
```

- 8 **Preview the AD modules.** These will be used from Lab 09 onward.

```
Get-WindowsFeature RSAT-AD-PowerShell, RSAT-ADDS, GPMC | Format-Table Name,
InstallState
```

EXPECTED OUTPUT

```
Computer Build UptimeHours IPv4      FreeC_GB FreeMemPct
-----
DC01      26100          3 10.10.10.10      41.2      52
SRV02     26100          3 10.10.10.21      29.8      61
VERBOSE: Saved to C:\Scripts\LabHealth-20260923-1415.csv
```

TROUBLESHOOTING

File cannot be loaded because running scripts is disabled	Execution policy was changed to Restricted. Set-ExecutionPolicy RemoteSigned -Scope LocalMachine.
Invoke-Command: Access is denied	Use SRV02\Administrator (local account) and confirm TrustedHosts includes the IP.
Format-Table output cannot be exported	Formatting destroys objects. Export before formatting, as the script does.

KNOWLEDGE CHECK

- Q1 Why does passing objects rather than text make filtering more reliable?
- Q2 What is the difference between one-to-one (Enter-PSSession) and one-to-many (Invoke-Command) remoting?
- Q3 Why should Format-* cmdlets appear only at the end of a pipeline?

SNAPSHOT ▸ L05-END-ps-report (DC01, SRV02)

Active Directory Core

What Active Directory Actually Is

Diagram the AD structure you will build across the rest of Part II.

OBJECTIVES

- Identify the AD DS components Greenfield will deploy
- Separate authentication from authorization in concrete scenarios
- Produce a reference diagram used through Lab 11

PREREQUISITES

No VM changes. Diagram tool (diagrams.net, Visio, or paper). Section B of this guide.

TASKS

- 1 **Draw the forest boundary.** One forest, one domain: greenfieldcorp.lab (NetBIOS GREENFIELD). Label it as the security boundary.
- 2 **Place the domain controllers.** DC01 (10.10.10.10) and DC02 (10.10.10.11) in site HQ-GRR, both DNS + Global Catalog. Mark DC01 as holder of all five FSMO roles initially.
- 3 **Add DNS.** Show the AD-integrated zones greenfieldcorp.lab and _msdcs.greenfieldcorp.lab, plus reverse zone 10.10.10.in-addr.arpa, and the forwarder to 1.1.1.1.
- 4 **Add member systems.** SRV01, SRV02 (member servers), CL01 (client). Draw arrows from each to DC01/DC02 labeled "DNS + Kerberos".
- 5 **Add the planned sites.** HQ-GRR (10.10.10.0/24) and BRANCH-DET (10.10.20.0/24) connected by site link HQ-BRANCH, cost 100, 15-minute interval.
- 6 **Annotate related services.** Mark which Greenfield needs now (AD DS), later (AD CS in Lab 21), and not at all in this course (AD LDS, AD FS, AD RMS). Add Entra ID outside the forest with a sync arrow (Lab 30).
- 7 **Write three authN vs authZ statements.** Example: "maria.lopez proves her identity to DC01 with Kerberos (authentication). The ENG share grants her Modify because she is in DL-ENG-Share-Modify (authorization)."
- 8 **Record schema facts from a running system later.** After Lab 09, confirm the schema version with the command below and add it to the diagram.

```
Get-ADObject (Get-ADRootDSE).schemaNamingContext -Properties objectVersion |  
Select-Object objectVersion
```

EXPECTED OUTPUT

A single-page diagram showing: forest/domain boundary, 2 DCs in HQ-GRR,
DNS zones, 3 members, 2 sites + 1 site link, Entra ID sync (future).
After Lab 09: objectVersion : 91

TROUBLESHOOTING

Diagram mirrors the org chart

Physical and logical structures are separate. Sites follow network topology; OUs (Lab 07) follow administration.

Unsure where Entra ID fits

Draw it outside the forest. It is a separate cloud directory that receives synchronized identities; it is not a domain controller.

KNOWLEDGE CHECK

- Q1 What problem would Greenfield have with 40 servers and no directory?
- Q2 Name the five AD server roles and one use for each.
- Q3 Why is DNS described as the nervous system of AD?

SNAPSHOT ▶ none – no VM changes in this lab

LAB 07

LESSON 7

Design the Logical Structure

Design the OU hierarchy for Greenfield Manufacturing on paper before building it.

OBJECTIVES

- Design an OU structure around administration and policy, not the org chart
- Choose a single-domain model and justify it
- Produce a build script ready to run after Lab 09

PREREQUISITES

Lab 06 diagram. No VM changes (the script is executed at the end of Lab 09).

TASKS

- 1 **List who administers what.** Greenfield: Tier 0 IT (4 people), Server team (Tier 1), Help desk (Tier 2), and departmental users in Engineering, Production, Finance, Sales, IT.
- 2 **List which settings differ by object type.** Workstations get lockdown GPOs; servers get server baselines; admin accounts get stricter password rules. These drive top-level OUs.

- 3 **Sketch the hierarchy.** Use section C of this guide as the target. Avoid a geography-first layout; locations change more often than administration models.
- 4 **Record the delegation plan.** Help desk: reset passwords in OU=Users. Server team: manage OU=Servers. Nobody but Tier 0: OU=Admin.
- 5 **Write C:\Scripts\New-GreenfieldOUs.ps1.** Save on your host now; copy to DC01 and run after promotion in Lab 09.

```
$root = (Get-ADDomain).DistinguishedName
New-ADOrganizationalUnit -Name Greenfield -Path $root -
ProtectedFromAccidentalDeletion $true
$gf = "OU=Greenfield,$root"
$tree = [ordered]@{
    'Admin'          = 'Tier0','Tier1','Tier2'
    'Users'          = 'Engineering','Production','Finance','Sales','IT'
    'Groups'         = 'Role','Resource'
    'Computers'      = 'Workstations','Servers'
    'ServiceAccounts' = @()
    'Disabled'       = @()
}
foreach ($top in $tree.Keys) {
    New-ADOrganizationalUnit -Name $top -Path $gf -ProtectedFromAccidentalDeletion
    $true
    foreach ($child in $tree[$top]) {
        New-ADOrganizationalUnit -Name $child -Path "OU=$top,$gf" -
        ProtectedFromAccidentalDeletion $true
    }
}
Get-ADOrganizationalUnit -SearchBase $gf -Filter * | Sort-Object DistinguishedName
| Select-Object DistinguishedName
```

- 6 **Peer-review the design.** Check: no OU used as a security group; default Computers/Users containers are not GPO targets (they are containers, not OUs); depth kept to three levels.

EXPECTED OUTPUT

```
Script saved. After running in Lab 09, 20 OUs are listed, e.g.
OU=Engineering,OU=Users,OU=Greenfield,DC=greenfieldcorp,DC=lab
OU=Tier0,OU=Admin,OU=Greenfield,DC=greenfieldcorp,DC=lab
```

TROUBLESHOOTING

Want an OU per plant location

Only add location OUs if administration or policy genuinely differs by location. Use the `/physicalDeliveryOfficeName` attributes otherwise.

Considering a second domain for Finance

Domains are not security boundaries; the forest is. Use OUs, delegation, and fine-grained password policies instead.

KNOWLEDGE CHECK

- Q1 Why is the forest, not the domain, the security boundary?
- Q2 Why can you link GPOs to OU=Users but not CN=Users?
- Q3 What role does the global catalog play in a single-domain forest?

SNAPSHOT ▸ none – no VM changes in this lab

LAB 08

LESSON 8

Sites, Replication, and the Physical Structure

Examine replication topology and create a second site with its own subnet.

OBJECTIVES

- Inspect the replication topology, KCC, and connection objects
- Create a site, subnet, and site link
- Locate Ntds.dit and SYSVOL on disk

PREREQUISITES

ORDER NOTE — this lab needs a domain. Complete Lab 09 first, then return here. Start from L09-END-dc01-promoted. DC02 (Lab 11) adds real connection objects; re-run step 2 then.

TASKS

1 Rename the default site. [DC01]

```
Get-ADReplicationSite -Filter * | Select-Object Name
Get-ADReplicationSite Default-First-Site-Name | Rename-ADObject -NewName HQ-GRR
```

2 Inspect topology and connections. With one DC there are no connection objects yet; the KCC runs every 15 minutes.

```
repadmin /showrepl
repadmin /kcc
Get-ADReplicationConnection -Filter * | Select-Object Name,
ReplicateFromDirectoryServer
Get-ADReplicationPartnerMetadata -Target DC01 -ErrorAction SilentlyContinue
```

3 Create subnets.

```
New-ADReplicationSubnet -Name '10.10.10.0/24' -Site HQ-GRR -Location 'Grand Rapids, MI'
New-ADReplicationSite -Name BRANCH-DET
New-ADReplicationSubnet -Name '10.10.20.0/24' -Site BRANCH-DET -Location 'Detroit, MI'
```

- 4 **Create the site link.** Replace DEFAULTIPSITELINK with an explicit link. Change notification is off inter-site by default.

```
New-ADReplicationSiteLink -Name HQ-BRANCH -SitesIncluded HQ-GRR,BRANCH-DET -Cost 100 -ReplicationFrequencyInMinutes 15 -InterSiteTransportProtocol IP
Get-ADReplicationSiteLink -Filter * | Select-Object Name, Cost, ReplicationFrequencyInMinutes, SitesIncluded
```

- 5 **Verify site assignment.**

```
nltest /dsgetsite
nltest /dsgetdc:greenfieldcorp.lab
```

- 6 **Locate the database and SYSVOL.**

```
Get-ItemProperty 'HKLM:\SYSTEM\CurrentControlSet\Services\NTDS\Parameters' |
    Select-Object 'DSA Database file','Database log files path'
Get-ChildItem C:\Windows\NTDS | Select-Object Name, Length
Get-SmbShare SYSVOL, NETLOGON
```

- 7 **Check the database page size.** New 2025 forests still use 8k pages until the optional feature is enabled (Lab 27).

```
Get-ADOptionalFeature -Filter * | Select-Object Name, EnabledScopes
ntdsutil "activate instance ntds" "files" "info" quit quit
```

EXPECTED OUTPUT

```
C:\> nltest /dsgetsite
HQ-GRR
Name          Cost ReplicationFrequencyInMinutes SitesIncluded
HQ-BRANCH     100                15 {CN=HQ-GRR..., CN=BRANCH-DET...}
DSA Database file : C:\Windows\NTDS\ntds.dit
```

TROUBLESHOOTING

**nltest /dsgetsite returns
ERROR_NO_SITENAME**

The DC IP is not in any subnet object. Create 10.10.10.0/24 and associate it with HQ-GRR, then restart Netlogon.

Rename-ADObject access denied

Run as a Domain/Enterprise Admin. Sites live in the Configuration partition.

KNOWLEDGE CHECK

- Q1 How does multi-master replication avoid a single primary?
- Q2 What does the KCC create, and how often does it run?
- Q3 When would you place an RODC instead of a writable DC?

SNAPSHOT ▸ L08-END-sites (DC01)

LAB 09

LESSON 9

Promote the First Domain Controller

Promote your first DC, then verify it with dcdiag and repadmin.

OBJECTIVES

- Complete the pre-promotion checklist
- Create the greenfieldcorp.lab forest with Install-ADDSForest
- Verify DNS, SYSVOL, services, and event logs after promotion
- Build the Lab 07 OU structure

PREREQUISITES

DC01 from L05-END-ps-report. Static IP 10.10.10.10, hostname DC01, correct time zone. Take L09-START snapshot first.

TASKS

1 Run the pre-promotion checklist. [DC01]

```
hostname
Get-NetIPConfiguration | Select-Object InterfaceAlias, IPv4Address,
IPv4DefaultGateway, DNSServer
(Get-NetIPInterface -InterfaceAlias LAN -AddressFamily IPv4).Dhcp    # must be
Disabled
w32tm /query /source
```

2 Install the AD DS role.

```
Install-WindowsFeature AD-Domain-Services -IncludeManagementTools
Get-WindowsFeature AD-Domain-Services, RSAT-AD-Tools, GPMC | Format-Table Name,
InstallState
```

3 Test the forest promotion. The forest is created at the Windows Server 2016 level on purpose, so that Lab 27 can practice raising it to the 2025 level (DomainLevel/ForestLevel 10).

```
$dsrm = Read-Host 'DSRM password' -AsSecureString
Test-ADDSForestInstallation -DomainName greenfieldcorp.lab -DomainNetbiosName
GREENFIELD `
  -ForestMode WinThreshold -DomainMode WinThreshold -InstallDns -
SafeModeAdministratorPassword $dsrm
```

- 4 **Record the DSRM password.** Store it in the lab password vault. It is required for authoritative restore (Lab 26) and cannot be recovered.

- 5 **Promote.** Ignore the DNS delegation warning; there is no parent zone for .lab. The server reboots automatically.

```
Install-ADDSForest -DomainName greenfieldcorp.lab -DomainNetbiosName GREENFIELD `
  -ForestMode WinThreshold -DomainMode WinThreshold -InstallDns `
  -DatabasePath C:\Windows\NTDS -LogPath C:\Windows\NTDS -SysvolPath
C:\Windows\SYSVOL `
  -SafeModeAdministratorPassword $dsrm -Force
```

- 6 **Log on as GREENFIELD\Administrator and fix DNS order.** Point to self first by IP, loopback second.

```
Set-DnsClientServerAddress -InterfaceAlias LAN -ServerAddresses
10.10.10.10,127.0.0.1
Get-Service NTDS, DNS, Netlogon, KDC, DFSR, W32Time | Format-Table Name, Status
```

- 7 **Verify with dcdiag and repadmin.**

```
dcdiag /v /c /e /q
dcdiag /test:dns /v
repadmin /showrepl
repadmin /replsummary
Get-SmbShare SYSVOL, NETLOGON
Get-ADDomain | Select-Object DNSRoot, NetBIOSName, DomainMode, PDCEmulator,
RIDMaster, InfrastructureMaster
Get-ADForest | Select-Object ForestMode, SchemaMaster, DomainNamingMaster,
GlobalCatalogs
```

- 8 **Check the event logs.**

```
Get-WinEvent -LogName 'Directory Service' -MaxEvents 30 | Where-Object Level -le 3
| Format-Table TimeCreated, Id, Message -Wrap
Get-WinEvent -LogName 'DFS Replication' -MaxEvents 10 | Where-Object Id -eq 4602
```

- 9 **Build the OU structure.** Run the Lab 07 script.

```
C:\Scripts\New-GreenfieldOUs.ps1
```

10 Now complete Lab 08. Take snapshot L09-END-dc01-promoted, then perform Lab 08 before continuing to Lab 10.

EXPECTED OUTPUT

```
dcdiag /q -> (no output = all tests passed)
DNSRoot      : greenfieldcorp.lab    NetBIOSName  : GREENFIELD
DomainMode   : Windows2016Domain    ForestMode   : Windows2016Forest
PDCEmulator: DC01.greenfieldcorp.lab
DFS Replication Event 4602: The DFS Replication service successfully initialized the SYSVOL replicated folder
```

TROUBLESHOOTING

dcdiag: failed test Connectivity / DNS	DNS client points somewhere other than 10.10.10.10. Fix the adapter, then ipconfig /registerdns and restart Netlogon.
SYSVOL share missing after promotion	Wait 5 minutes for DFSR event 4602. If absent, check HKLM\...\Netlogon\Parameters SysvolReady = 1.
Promotion blocked: local Administrator password does not meet requirements	It becomes the domain Administrator password. Set a complex one with net user Administrator *.

KNOWLEDGE CHECK

Q1 What three things does promotion create on disk?

Q2 Why is the DSRM password different from the domain Administrator password?

Q3 Which FSMO roles does DC01 hold after promotion, and why?

SNAPSHOT ▶ L09-END-dc01-promoted (DC01)

DNS, the Foundation Underneath Everything

Inspect the SRV records AD created, add a reverse zone, and configure a forwarder.

OBJECTIVES	• Read the SRV records clients use to locate DCs • Create an AD-integrated reverse lookup zone • Configure forwarders and scavenging • Troubleshoot with nslookup and Resolve-DnsName
------------	---

PREREQUISITES	DC01 from L08-END-sites. SRV02 running.
---------------	---

TASKS

1 List zones. [DC01]

```
Get-DnsServerZone | Format-Table ZoneName, ZoneType, IsDsIntegrated,  
ReplicationScope, IsReverseLookupZone
```

2 Read the DC locator SRV records.

```
Resolve-DnsName -Type SRV _ldap._tcp.dc._msdcs.greenfieldcorp.lab  
Resolve-DnsName -Type SRV _kerberos._tcp.greenfieldcorp.lab  
Resolve-DnsName -Type SRV _ldap._tcp.HQ-GRR._sites.dc._msdcs.greenfieldcorp.lab  
Resolve-DnsName -Type SRV _gc._tcp.greenfieldcorp.lab  
Get-DnsServerResourceRecord -ZoneName _msdcs.greenfieldcorp.lab -RRType CNAME
```

3 Read them the old way.

```
nslookup  
> set type=srv  
> _ldap._tcp.dc._msdcs.greenfieldcorp.lab  
> exit
```

4 Create the reverse zone.

```
Add-DnsServerPrimaryZone -NetworkId '10.10.10.0/24' -ReplicationScope Forest -  
DynamicUpdate Secure  
Add-DnsServerPrimaryZone -NetworkId '10.10.20.0/24' -ReplicationScope Forest -  
DynamicUpdate Secure  
ipconfig /registerdns  
Add-DnsServerResourceRecordPtr -ZoneName 10.10.10.in-addr.arpa -Name 21 -  
PtrDomainName SRV02.greenfieldcorp.lab
```

5 Configure forwarders. Replace root hints with explicit forwarders for faster external resolution.

```
Set-DnsServerForwarder -IPAddress 1.1.1.1, 9.9.9.9 -UseRootHint $true  
Get-DnsServerForwarder  
Resolve-DnsName www.microsoft.com -Server 10.10.10.10
```

6 Add a conditional forwarder (example). Simulates a partner domain. It will not resolve; that is expected.

```
Add-DnsServerConditionalForwarderZone -Name partner.example -MasterServers  
10.10.10.250 -ReplicationScope Forest
```

7 Enable aging and scavenging.

```
Set-DnsServerScavenging -ScavengingState $true -ScavengingInterval 7.00:00:00 -
ApplyOnAllZones
Set-DnsServerZoneAging greenfieldcorp.lab -Aging $true -NoRefreshInterval
7.00:00:00 -RefreshInterval 7.00:00:00
Get-DnsServerScavenging
```

8 Test from a member. [SRV02]

```
Resolve-DnsName DC01.greenfieldcorp.lab
Resolve-DnsName 10.10.10.10
nltest /dsgetdc:greenfieldcorp.lab
```

EXPECTED OUTPUT

```
Name                                Type TTL  Section NameTarget
Priority Weight Port
_ldap._tcp.dc._msdcs.greenfieldcorp.lab  SRV  600  Answer  dc01.greenfieldcorp.lab 0
100      389
10.10.10.10.in-addr.arpa PTR    DC01.greenfieldcorp.lab
IPAddress : {1.1.1.1, 9.9.9.9} UseRootHint : True
```

TROUBLESHOOTING

SRV records missing	Restart Netlogon (Restart-Service Netlogon) to re-register from %SystemRoot%\System32\Config\netlogon.dns, then dcdiag /test:registerindns /dnsdomain:greenfieldcorp.lab.
External names fail	Check outbound UDP/TCP 53 through GFM-NAT and the forwarder list. Test with Resolve-DnsName -Server 1.1.1.1.
PTR records not created automatically	Reverse zone was created after registration. Run ipconfig /registerdns on each host.

KNOWLEDGE CHECK

- Q1 Which SRV record does a client query first to find a DC in its own site?
- Q2 What is the advantage of an AD-integrated zone over a file-backed primary?
- Q3 What problem do aging and scavenging solve, and what is the risk of aggressive intervals?

SNAPSHOT ▸ L10-END-dns (DC01, SRV02)

FSMO Roles and the Global Catalog

Locate all FSMO holders, transfer a role to a second DC, and transfer it back.

OBJECTIVES

- Build and promote DC02 as an additional DC
- Locate FSMO role holders with netdom and PowerShell
- Transfer roles gracefully and understand seizing
- Verify GC status and PDC time source

PREREQUISITES

DC01 from L10-END-dns. TPL-WS2025 template for DC02.

TASKS

1 Clone and configure DC02. [host, then DC02]

```
VBoxManage clonevm TPL-WS2025 --snapshot "L02-END-template-sysprepped" --options link --name DC02 --register
VBoxManage modifyvm DC02 --memory 3072
VBoxManage startvm DC02
# inside DC02 after OOBE:
New-NetIPAddress -InterfaceAlias Ethernet -IPAddress 10.10.10.11 -PrefixLength 24 -DefaultGateway 10.10.10.1
Set-DnsClientServerAddress -InterfaceAlias Ethernet -ServerAddresses 10.10.10.10
Rename-Computer DC02 -Restart
```

2 Join and promote DC02. [DC02]

```
Add-Computer -DomainName greenfieldcorp.lab -Credential GREENFIELD\Administrator -Restart
# after reboot, as GREENFIELD\Administrator:
Install-WindowsFeature AD-Domain-Services -IncludeManagementTools
Install-ADDSDomainController -DomainName greenfieldcorp.lab -InstallDns -SiteName HQ-GRR `
    -Credential (Get-Credential GREENFIELD\Administrator) `
    -SafeModeAdministratorPassword (Read-Host DSRM -AsSecureString) -Force
```

3 Fix DNS on both DCs. Each DC points to its partner first, itself second.

```
Set-DnsClientServerAddress -InterfaceAlias Ethernet -ServerAddresses 10.10.10.10,10.10.10.11 # DC02
Set-DnsClientServerAddress -InterfaceAlias LAN -ServerAddresses 10.10.10.11,10.10.10.10 # DC01
repadmin /replsummary
repadmin /showrepl DC02
```

4 Locate role holders. [DC01]


```
netdom query fsmo
Get-ADDomain | Select-Object PDCEmulator, RIDMaster, InfrastructureMaster
Get-ADForest | Select-Object SchemaMaster, DomainNamingMaster
Get-ADDomainController -Filter * | Select-Object Name, IsGlobalCatalog,
OperationMasterRoles
```

5 Transfer the RID Master and PDC Emulator to DC02.

```
Move-ADDirectoryServerOperationMasterRole -Identity DC02 -OperationMasterRole
RIDMaster, PDCEmulator -Confirm:$false
netdom query fsmo
```

6 Check time hierarchy after the PDC move. Only the PDC Emulator of the forest root should sync externally; others use NT5DS.

```
w32tm /query /source           # on DC01: should be DC02
w32tm /monitor /domain:greenfieldcorp.lab
```

7 Transfer the roles back.

```
Move-ADDirectoryServerOperationMasterRole -Identity DC01 -OperationMasterRole
RIDMaster, PDCEmulator -Confirm:$false
netdom query fsmo
```

8 Configure the PDC time source. [DC01]

```
w32tm /config /manualpeerlist:"time.windows.com,0x8 pool.ntp.org,0x8"
/syncfromflags:manual /reliable:yes /update
Restart-Service w32time; w32tm /resync /rediscover
w32tm /query /status
```

9 Understand seizing (read only — do not run). Seize only when the holder is permanently gone and will never return to the network: Move-ADDirectoryServerOperationMasterRole -Identity DC02 -OperationMasterRole RIDMaster -Force. A seized RID Master that comes back online can issue duplicate RID pools.

EXPECTED OUTPUT

```
Schema master           DC01.greenfieldcorp.lab
Domain naming master    DC01.greenfieldcorp.lab
PDC                     DC01.greenfieldcorp.lab
RID pool manager        DC01.greenfieldcorp.lab
Infrastructure master    DC01.greenfieldcorp.lab
Name IsGlobalCatalog
DC01                    True
DC02                    True
```

TROUBLESHOOTING

Install-ADDSDomainController: cannot contact domain DC02 DNS must point at 10.10.10.10 before joining. Test with `nltest /dsgetdc:greenfieldcorp.lab`.

Move-ADDirectoryServerOperationMasterRole: Schema/Enterprise Admins. Use GREENFIELD\Administrator. access denied Schema and Domain Naming roles require Schema/Enterprise Admins. Use GREENFIELD\Administrator.

repadmin shows 1722 RPC server unavailable Firewall profile on DC02 is not Domain. Restart NlaSvc or reboot DC02.

KNOWLEDGE CHECK

- Q1 Why is the PDC Emulator the most operationally important FSMO role?
- Q2 What happens if the Infrastructure Master is on a GC in a multi-domain forest where not all DCs are GCs?
- Q3 What is the difference between transferring and seizing, and why is seizing dangerous?

SNAPSHOT ▸ L11-END-dc02-fsmo (DC01, DC02)

Identity, Policy, and Access

Users, Groups, and Computers

Bulk-create a department of users and groups from a CSV and apply correct nesting.

OBJECTIVES

- Join SRV01, SRV02, and CL01 to the domain
- Bulk-create users from CSV with PowerShell
- Implement AGDLP nesting for a resource
- Find stale accounts and apply disable-before-delete

PREREQUISITES

DC01, DC02 from L11-END-dc02-fsmo. SRV01 cloned from the template (10.10.10.20). CL01 installed from the Windows 11 Enterprise evaluation ISO (static 10.10.10.50, DNS 10.10.10.10).

TASKS

- 1 **Pre-stage computer accounts. [DC01]** Pre-staging places servers in the right OU. 2025 join hardening blocks reuse of an existing computer account unless the joining user created it or is a Domain Admin, and no longer accepts default (name-based) passwords.

```
$srv = 'OU=Servers,OU=Computers,OU=Greenfield,DC=greenfieldcorp,DC=lab'
$wks = 'OU=Workstations,OU=Computers,OU=Greenfield,DC=greenfieldcorp,DC=lab'
redircmp $wks
New-ADComputer SRV01 -Path $srv; New-ADComputer SRV02 -Path $srv
```

- 2 **Join members. [SRV01, SRV02, CL01]**

```
Add-Computer -DomainName greenfieldcorp.lab -Credential GREENFIELD\Administrator -
Restart
# SRV01 & SRV02: remove workgroup TrustedHosts on DC01 afterward
Clear-Item WSMAN:\localhost\Client\TrustedHosts -Force # on DC01
```

- 3 **Verify the secure channel. [SRV02]**

```
Test-ComputerSecureChannel -Verbose
nltest /sc_verify:greenfieldcorp.lab
```

- 4 **Create the CSV. [DC01]** Save as C:\Scripts\engineering.csv.

```

First,Last,Title,Role
Maria,Lopez,Design Engineer,Designers
James,Okafor,Design Engineer,Designers
Priya,Nair,Process Engineer,Process
Tom,Becker,Process Engineer,Process
Lena,Hoffmann,CAD Technician,Designers
Sam,Reyes,Engineering Manager,Managers

```

5 Create role and resource groups.

```

$role = 'OU=Role,OU=Groups,OU=Greenfield,DC=greenfieldcorp,DC=lab'
$res = 'OU=Resource,OU=Groups,OU=Greenfield,DC=greenfieldcorp,DC=lab'
'Designers','Process','Managers' | ForEach-Object {
    New-ADGroup "GG-ENG-$_" -GroupScope Global -GroupCategory Security -Path $role }
New-ADGroup UG-AllEngineering -GroupScope Universal -Path $role
New-ADGroup DL-ENG-Share-Modify -GroupScope DomainLocal -Path $res
New-ADGroup DL-ENG-Share-Read -GroupScope DomainLocal -Path $res

```

6 Bulk-create users.

```

$ou = 'OU=Engineering,OU=Users,OU=Greenfield,DC=greenfieldcorp,DC=lab'
$pw = Read-Host 'Initial password' -AsSecureString
Import-Csv C:\Scripts\engineering.csv | ForEach-Object {
    $sam = ("$(_.First).$(_.Last)").ToLower()
    New-ADUser -Name "$($_.First) $_.Last" -GivenName $_.First -Surname $_.Last `
        -SamAccountName $sam -UserPrincipalName "$sam@greenfieldcorp.lab" -Title
    $_.Title `
        -Department Engineering -Path $ou -AccountPassword $pw -Enabled $true -
    ChangePasswordAtLogon $true
    Add-ADGroupMember "GG-ENG-$(_.Role)" -Members $sam
}

```

7 Apply AGDLP nesting. Accounts → Global → (Universal) → Domain Local → Permission.

```

Add-ADGroupMember UG-AllEngineering -Members GG-ENG-Designers, GG-ENG-Process, GG-
ENG-Managers
Add-ADGroupMember DL-ENG-Share-Modify -Members UG-AllEngineering
Get-ADGroupMember DL-ENG-Share-Modify -Recursive | Select-Object SamAccountName

```

8 Practice lifecycle operations.

```
Disable-ADAccount tom.becker
Move-ADObject (Get-ADUser tom.becker).DistinguishedName -TargetPath
'OU=Disabled,OU=Greenfield,DC=greenfieldcorp,DC=lab'
Search-ADAccount -LockedOut | Unlock-ADAccount
Search-ADAccount -AccountInactive -TimeSpan 90.00:00:00 -UsersOnly | Select-Object
Name, LastLogonDate
Search-ADAccount -AccountInactive -TimeSpan 90.00:00:00 -ComputersOnly
```

- 9 **Log on to CL01 as maria.lopez.** Confirm the forced password change and that `whoami /groups` shows the nested groups.

```
whoami /upn
whoami /groups | findstr /i "ENG"
```

EXPECTED OUTPUT

```
Test-ComputerSecureChannel -> True
SamAccountName
-----
maria.lopez
james.okafor
priya.nair
lena.hoffmann
sam.reyes
(tom.becker absent: disabled and moved)
```

TROUBLESHOOTING

Join fails: "an account with the same name exists"	2025 blocks takeover of pre-created accounts unless the joining user is the account owner or in the allowlist. Join with Domain Admin, or delete and retry.
New-ADUser: password does not meet requirements	Default domain policy requires 7+ characters and complexity. Use the lab password.
whoami /groups missing new groups	Group membership is evaluated at logon. Sign out and back in (or klist purge for network access).

KNOWLEDGE CHECK

- Q1 When would a user sign in with UPN versus SamAccountName?
- Q2 Explain AGDLP in one sentence, and why permissions go on Domain Local groups.
- Q3 Why disable, then delete after a waiting period?

SNAPSHOT ▸ L12-END-users-groups (DC01, DC02, SRV01, SRV02, CL01)

Group Policy Fundamentals

Build and link GPOs for a desktop lockdown and verify precedence with gpresult.

OBJECTIVES

- Create, link, and name GPOs to the standard
- Observe LSDOU precedence with a deliberate conflict
- Use block inheritance, enforced links, and security filtering
- Verify results with gpresult

PREREQUISITES

L12-END-users-groups. DC01, CL01 running; maria.lopez has logged on to CL01 once.

TASKS

- 1 **Create the lockdown GPO. [DC01]** Prohibits Control Panel and sets a screen-lock timeout for users in OU=Users.

```
$users = 'OU=Users,OU=Greenfield,DC=greenfieldcorp,DC=lab'
New-GPO -Name 'USR-U-DesktopLockdown-v1' -Comment 'Lab 13 desktop lockdown'
Set-GPRegistryValue -Name 'USR-U-DesktopLockdown-v1' -Key
'HKCU\Software\Microsoft\Windows\CurrentVersion\Policies\Explorer' -ValueName
NoControlPanel -Type DWord -Value 1
Set-GPRegistryValue -Name 'USR-U-DesktopLockdown-v1' -Key
'HKCU\Software\Policies\Microsoft\Windows\Control Panel\Desktop' -ValueName
ScreenSaveTimeOut -Type String -Value '600'
Set-GPRegistryValue -Name 'USR-U-DesktopLockdown-v1' -Key
'HKCU\Software\Policies\Microsoft\Windows\Control Panel\Desktop' -ValueName
ScreenSaverIsSecure -Type String -Value '1'
New-GPLink -Name 'USR-U-DesktopLockdown-v1' -Target $users
```

- 2 **Create a conflicting domain-level GPO.** Sets a 15-minute timeout at the domain. By LSDOU, the OU link (applied last) wins.

```
New-GPO -Name 'DOM-U-ScreenSaver-v1' | New-GPLink -Target
'DC=greenfieldcorp,DC=lab'
Set-GPRegistryValue -Name 'DOM-U-ScreenSaver-v1' -Key
'HKCU\Software\Policies\Microsoft\Windows\Control Panel\Desktop' -ValueName
ScreenSaveTimeOut -Type String -Value '900'
```

- 3 **Apply and verify on the client. [CL01 as maria.lopez]**

```
gpupdate /force
gpresult /r /scope user
gpresult /h $env:USERPROFILE\Desktop\gp.html
reg query "HKCU\Software\Policies\Microsoft\Windows\Control Panel\Desktop" /v
ScreenSaveTimeOut
```

- 4 **Enforce the domain link. [DC01]** An enforced link wins over lower links and ignores block inheritance.

```
Set-GPLink -Name 'DOM-U-ScreenSaver-v1' -Target 'DC=greenfieldcorp,DC=lab' -
Enforced Yes
Get-GPInheritance -Target $users | Select-Object -ExpandProperty InheritedGpoLinks
```

- 5 **Re-test, then remove enforcement.** On CL01: gpupdate /force; value should read 900. Then revert.

```
Set-GPLink -Name 'DOM-U-ScreenSaver-v1' -Target 'DC=greenfieldcorp,DC=lab' -
Enforced No
```

- 6 **Block inheritance on IT users.**

```
Set-GPInheritance -Target 'OU=IT,OU=Users,OU=Greenfield,DC=greenfieldcorp,DC=lab'
-IsBlocked Yes
```

- 7 **Security-filter the lockdown to Engineering only.** Remove Authenticated Users Apply, keep Read (required since MS16-072), and add Apply for the Engineering group.

```
$g='USR-U-DesktopLockdown-v1'
Set-GPPermission -Name $g -TargetName 'Authenticated Users' -TargetType Group -
PermissionLevel GpoRead -Replace
Set-GPPermission -Name $g -TargetName 'UG-AllEngineering' -TargetType Group -
PermissionLevel GpoApply
Get-GPPermission -Name $g -All | Select-Object Trustee, Permission
```

- 8 **Add a WMI filter (optional).** GPMC > WMI Filters > New: SELECT * FROM Win32_OperatingSystem WHERE ProductType = 1 (workstations only). Link it to a computer GPO.

- 9 **Back up both GPOs.**

```
New-Item C:\GPOBackup -ItemType Directory -Force
Backup-GPO -All -Path C:\GPOBackup
```

EXPECTED OUTPUT

Applied Group Policy Objects

```
-----  
    USR-U-DesktopLockdown-v1  
    DOM-U-ScreenSaver-v1  
ScreenSaveTimeout    REG_SZ    600    (900 while domain link is Enforced)
```

TROUBLESHOOTING

GPO listed under "Filtered out (Security)"

The user is not in a group with Apply, or Authenticated Users lost Read. Check Get-GPPermission.

Settings do not change after gpupdate

Some user settings need a logoff. Run gpupdate /force /logoff.

gpreresult: user does not have RSoP data

Run as the user who logged on, or use /user GREENFIELD\maria.lopez from an elevated prompt.

KNOWLEDGE CHECK

- Q1 State LSDOU and which level wins a conflict by default.
- Q2 Where do the GPC and GPT live, and what does each contain?
- Q3 Why is one giant GPO an operational risk?

SNAPSHOT ▸ L13-END-gpo-basics (DC01, CL01)

LAB 14

LESSON 14

Group Policy in Depth

Implement folder redirection and a fine-grained password policy for privileged accounts.

OBJECTIVES

- Create the ADMX central store
- Deploy folder redirection to a secured share
- Use Group Policy Preferences with item-level targeting
- Create a Password Settings Object for Tier 0 admins

PREREQUISITES

L13-END-gpo-basics. SRV01 has a D: data disk (add a 40 GB VDI and initialize it).

TASKS

1 Create the central store. [DC01]

```
$cs='\\greenfieldcorp.lab\SYSTEM\greenfieldcorp.lab\Policies\PolicyDefinitions'  
Copy-Item C:\Windows\PolicyDefinitions $cs -Recurse  
Get-ChildItem $cs -Filter *.admx | Measure-Object
```


- 2 **Create the redirection share. [SRV01]** Users get Create Folder on the root only; CREATOR OWNER gets Full on subfolders.

```
New-Item D:\Redirect -ItemType Directory
icacls D:\Redirect /inheritance:r
icacls D:\Redirect /grant "SYSTEM:(OI)(CI)F" "Administrators:(OI)(CI)F" "CREATOR
OWNER:(OI)(CI)(IO)F"
icacls D:\Redirect /grant "GREENFIELD\Domain Users:(RD,AD,X,RA)"
New-SmbShare -Name Redirect$ -Path D:\Redirect -FullAccess 'Authenticated Users' -
FolderEnumerationMode AccessBased
```

- 3 **Configure folder redirection. [DC01, GPMC]** New GPO USR-U-FolderRedirect-v1 linked to OU=Engineering. User Config > Policies > Windows Settings > Folder Redirection > Documents > Basic, "Create a folder for each user", root \\SRV01\Redirect\$. Settings tab: clear "Grant the user exclusive rights".
- 4 **Add a mapped drive preference with targeting.** User Config > Preferences > Drive Maps > New: Update, \\SRV01\Engineering, letter E:. Common > Item-level targeting > Security Group = GREENFIELD\UG-AllEngineering. (Share is created in Lab 24; the map will appear then.)
- 5 **Verify redirection. [CL01 as maria.lopez]**

```
gpupdate /force /logoff
# after logon:
[Environment]::GetFolderPath('MyDocuments')
```

- 6 **Create a Tier 0 admin and group. [DC01]**

```
$t0='OU=Tier0,OU=Admin,OU=Greenfield,DC=greenfieldcorp,DC=lab'
New-ADGroup GG-T0-Admins -GroupScope Global -Path $t0
New-ADUser adm-t0-maria.lopez -UserPrincipalName adm-t0-
maria.lopez@greenfieldcorp.lab -Path $t0 `
    -AccountPassword (Read-Host -AsSecureString) -Enabled $true
Add-ADGroupMember GG-T0-Admins adm-t0-maria.lopez
Add-ADGroupMember 'Domain Admins' adm-t0-maria.lopez
```

- 7 **Create the fine-grained password policy.**

```
New-ADFineGrainedPasswordPolicy -Name PS0-Tier0 -Precedence 10 -MinPasswordLength
16 `
    -ComplexityEnabled $true -PasswordHistoryCount 24 -MaxPasswordAge 90.00:00:00 -
MinPasswordAge 1.00:00:00 `
    -LockoutThreshold 5 -LockoutDuration 00:30:00 -LockoutObservationWindow 00:30:00
-ReversibleEncryptionEnabled $false
Add-ADFineGrainedPasswordPolicySubject PS0-Tier0 -Subjects GG-T0-Admins
Get-ADUserResultantPasswordPolicy adm-t0-maria.lopez
```

- 8 **Test the PSO.** A 12-character password must fail for the Tier 0 account but succeed for a standard user.

```
Set-ADAccountPassword adm-t0-maria.lopez -Reset -NewPassword (ConvertTo-SecureString 'Short.Pass12' -AsPlainText -Force)
```

- 9 **Review the operational log. [CL01]**

```
Get-WinEvent -LogName 'Microsoft-Windows-GroupPolicy/Operational' -MaxEvents 20 | Format-Table TimeCreated, Id, Message -Wrap
```

EXPECTED OUTPUT

```
MyDocuments -> \\SRV01\Redirect$\maria.lopez\Documents
Name          : PSO-Tier0
MinPasswordLength : 16
Set-ADAccountPassword : The password does not meet the length, complexity, or history requirement of the domain.
```

TROUBLESHOOTING

Folder Redirection event 502: failed to apply	Share or NTFS permissions wrong, or "exclusive rights" left on. Re-check icacs and the policy Settings tab.
Get-ADUserResultantPasswordPolicy returns nothing	PSO applied to a group the user is not directly or globally a member of. PSOs apply only to users and global security groups.
ADMX editor shows "Administrative Templates: Policy definitions retrieved from local computer"	The central store path is misspelled. It must be ...\\Policies\\PolicyDefinitions.

KNOWLEDGE CHECK

- Q1 What is the difference between a policy and a preference in how settings are removed?
- Q2 Why must the domain password policy be set in a domain-linked GPO while PSOs target groups?
- Q3 When is item-level targeting preferable to a separate GPO?

SNAPSHOT ▸ L14-END-gpo-depth (DC01, SRV01, CL01)

Service Accounts and Managed Identities

Create a gMSA and a dMSA, install them on a member server, and migrate a service.

OBJECTIVES

- Create the KDS root key
- Create and install a gMSA on SRV02
- Create a dMSA and migrate a legacy svc- account to it
- Inventory service accounts and SPNs

PREREQUISITES

L14-END-gpo-depth. SRV02 (Server Core) running. dMSA requires a Windows Server 2025 DC (DC01/DC02).

TASKS

- 1 **Create the KDS root key. [DC01]** In production wait 10 hours for replication. In a lab, back-date it.

```
Add-KdsRootKey -EffectiveTime ((Get-Date).AddHours(-10))
Get-KdsRootKey
```

- 2 **Create a host group and gMSA.**

```
$sa='OU=ServiceAccounts,OU=Greenfield,DC=greenfieldcorp,DC=lab'
New-ADGroup GG-gMSA-WebApp-Hosts -GroupScope Global -Path $sa
Add-ADGroupMember GG-gMSA-WebApp-Hosts -Members 'SRV02$'
New-ADServiceAccount gmsa-webapp -DNSHostName gmsa-webapp.greenfieldcorp.lab -Path $sa `
    -PrincipalsAllowedToRetrieveManagedPassword GG-gMSA-WebApp-Hosts -
ManagedPasswordIntervalInDays 30
```

- 3 **Install and test on SRV02. [SRV02]** Reboot first so SRV02 picks up the new group membership.

```
Restart-Computer
Install-WindowsFeature RSAT-AD-PowerShell
Install-ADServiceAccount gmsa-webapp
Test-ADServiceAccount gmsa-webapp
```

- 4 **Create a legacy service and account. [DC01 then SRV02]** Simulates the inherited svc-account pattern.

```
New-ADUser svc-legacyapp -Path $sa -AccountPassword (Read-Host -AsSecureString) -
Enabled $true -PasswordNeverExpires $true
Set-ADUser svc-legacyapp -ServicePrincipalNames
@{Add='HTTP/srv02.greenfieldcorp.lab'}
# SRV02:
New-Service -Name LegacyApp -BinaryPathName 'C:\Windows\System32\svchost.exe -k
netsvcs' -StartupType Manual `
-Credential (Get-Credential GREENFIELD\svc-legacyapp)
```

- 5 **Run a service as the gMSA. [SRV02]** The trailing \$ is required; the password is blank.

```
sc.exe config LegacyApp obj= "GREENFIELD\gmsa-webapp$" password= ""
Get-CimInstance Win32_Service -Filter "Name='LegacyApp'" | Select-Object Name,
StartName
```

- 6 **Create a dMSA. [DC01]**

```
New-ADServiceAccount -Name dmsa-legacyapp -DNSHostName dmsa-
legacyapp.greenfieldcorp.lab `
-CreateDelegatedServiceAccount -KerberosEncryptionType AES256 -Path $sa
Set-ADServiceAccount dmsa-legacyapp -PrincipalsAllowedToRetrieveManagedPassword
'SRV02$'
```

- 7 **Start migration from svc-legacyapp to the dMSA.** The dMSA learns which machines use the legacy account during the migration window.

```
Start-ADServiceAccountMigration -Identity dmsa-legacyapp -SupersededAccount svc-
legacyapp
Get-ADServiceAccount dmsa-legacyapp -Properties msDS-DelegatedMSAState, msDS-
ManagedAccountPrecededByLink
```

- 8 **Complete migration.** After the service has restarted on SRV02 under the legacy account at least once, complete. The legacy account is disabled for authentication and its tickets are redirected to the dMSA.

```
Restart-Service LegacyApp -ErrorAction SilentlyContinue # on SRV02 (if
configured with legacy account)
Complete-ADServiceAccountMigration -Identity dmsa-legacyapp -SupersededAccount
svc-legacyapp
Get-ADServiceAccount dmsa-legacyapp -Properties msDS-DelegatedMSAState
```

- 9 **Inventory service account sprawl.**

```
Get-ADUser -Filter 'ServicePrincipalName -like "*" ' -Properties
ServicePrincipalName, PasswordLastSet, PasswordNeverExpires |
    Select-Object SamAccountName, PasswordLastSet, PasswordNeverExpires,
    @{n='SPNs';e={$_.ServicePrincipalName -join ';'}}
```

```
Get-ADServiceAccount -Filter * | Select-Object Name, ObjectClass
```

EXPECTED OUTPUT

```
Test-ADServiceAccount gmsa-webapp -> True
StartName : GREENFIELD\gmsa-webapp$
msDS-DelegatedMSAState : 1 (migration in progress) -> 2 (migration completed)
Name          ObjectClass
gmsa-webapp    msDS-GroupManagedServiceAccount
dmsa-legacyapp msDS-DelegatedManagedServiceAccount
```

TROUBLESHOOTING

Test-ADServiceAccount returns False	SRV02 has not refreshed its Kerberos ticket since joining the group. Reboot SRV02 or run <code>klist -li 0x3e7 purge</code> .
New-ADServiceAccount: key does not exist	KDS root key not yet effective. Confirm EffectiveTime is in the past on all DCs.
dMSA parameters not recognized	RSAT on the machine running the command must be from Windows Server 2025.

KNOWLEDGE CHECK

- Q1 Why are static-password svc- accounts attractive to Kerberoasting?
- Q2 What is the key difference between how gMSA and dMSA passwords are tied to machines?
- Q3 What does msDS-ManagedAccountPrecededByLink record, and why does it matter for security (preview of Lesson 19)?

SNAPSHOT ▸ L15-END-managed-accounts (DC01, SRV02)

Delegation of Administration

Delegate help desk password-reset rights to an OU and verify the boundary holds.

OBJECTIVES	• Delegate reset password and unlock rights with dscls and the wizard • Read the resulting ACEs • Prove the delegation does not extend outside the OU or to admins • Add a Tier 0 account to Protected Users
------------	--

PREREQUISITES	L15-END-managed-accounts. DC01, CL01.
---------------	---------------------------------------

TASKS

1 Create the help desk group and a technician. [DC01]

```
$t2='OU=Tier2,OU=Admin,OU=Greenfield,DC=greenfieldcorp,DC=lab'  
New-ADGroup GG-T2-HelpDesk -GroupScope Global -Path $t2  
New-ADUser adm-t2-sam.reyes -UserPrincipalName adm-t2-sam.reyes@greenfieldcorp.lab  
-Path $t2 -AccountPassword (Read-Host -AsSecureString) -Enabled $true  
Add-ADGroupMember GG-T2-HelpDesk adm-t2-sam.reyes
```

2 Delegate with dscls. Grants reset password, force change at next logon, and unlock on user objects in OU=Users only.

```
$ou='OU=Users,OU=Greenfield,DC=greenfieldcorp,DC=lab'  
dscls $ou /I:S /G "GREENFIELD\GG-T2-HelpDesk:CA;Reset Password;user"  
dscls $ou /I:S /G "GREENFIELD\GG-T2-HelpDesk:RPWP;pwdLastSet;user"  
dscls $ou /I:S /G "GREENFIELD\GG-T2-HelpDesk:RPWP;lockoutTime;user"
```

3 Read the ACL back.

```
(Get-Acl "AD:$ou").Access | Where-Object IdentityReference -like '*HelpDesk*' |  
Select-Object ActiveDirectoryRights, ObjectType, InheritedObjectType,  
InheritanceType
```

4 Test inside the boundary. [CL01 as adm-t2-sam.reyes] Install RSAT: Add- WindowsCapability -Online -Name Rsat.ActiveDirectory.DS-LDS.Tools~~~~0.0.1.0

```
Set-ADAccountPassword priya.nair -Reset -NewPassword (Read-Host -AsSecureString)  
Set-ADUser priya.nair -ChangePasswordAtLogon $true  
Unlock-ADAccount priya.nair
```

5 Test outside the boundary. Both commands must fail with Access is denied.

```
Set-ADAccountPassword adm-t0-maria.lopez -Reset -NewPassword (Read-Host -
AsSecureString)
Set-ADUser priya.nair -Title 'Changed'
```

- 6 **Understand AdminSDHolder.** Members of protected groups have inheritance disabled every 60 minutes by SDProp, so OU delegations never reach them.

```
Get-ADUser adm-t0-maria.lopez -Properties adminCount | Select-Object Name,
adminCount
```

- 7 **Add Tier 0 to Protected Users. [DC01]** Blocks NTLM, DES/RC4, delegation, and cached credentials for members. Never add service or computer accounts.

```
Add-ADGroupMember 'Protected Users' -Members adm-t0-maria.lopez
```

- 8 **Check remote SAM restrictions.** 2025 restricts remote SAM enumeration to Administrators by default.

```
Get-ItemProperty HKLM:\SYSTEM\CurrentControlSet\Control\Lsa -Name
RestrictRemoteSAM -ErrorAction SilentlyContinue
```

- 9 **Document the delegation.** Record in C:\Scripts\delegations.csv: OU, trustee, rights, date, approver. Export the current state:

```
(Get-Acl "AD:$ou").Access | Where-Object { -not $_.IsInherited -and
$_ .IdentityReference -notlike 'NT AUTHORITY*' } |
Export-Csv C:\Scripts\delegations-$(Get-Date -Format yyyyMMdd).csv -
NoTypeInfo
```

EXPECTED OUTPUT

```
ActiveDirectoryRights ObjectType InheritedObjectType
ExtendedRight 00299570-246d-11d0-a768-00aa006e0529 bf967aba-0de6-11d0-a285-
00aa003049e2
Set-ADAccountPassword adm-t0-maria.lopez : Access is denied
adminCount : 1
```

TROUBLESHOOTING

Help desk can reset some Tier 0 accounts

Those accounts are not in a protected group and sit in the delegated OU. Move all admin accounts to OU=Admin.

dsacIs: The parameter is incorrect

Quote the whole grant string; right names are case-sensitive (Reset Password).

KNOWLEDGE CHECK

- Q1 What does the Delegation of Control wizard actually write?
- Q2 Why does delegation on OU=Users not affect Domain Admins stored there?
- Q3 Describe the clean-source principle in the tier model.

SNAPSHOT ▸ L16-END-delegation (DC01, CL01)

LAB 17

LESSON 17

Authentication Protocols: Kerberos and NTLM

Register an SPN, configure constrained delegation, and trace a Kerberos ticket with klist.

OBJECTIVES

- Register and validate SPNs, including duplicate detection
- Trace AS, TGS, and service tickets with klist
- Configure Kerberos constrained and resource-based constrained delegation
- Audit NTLM usage

PREREQUISITES

L16-END-delegation. DC01, SRV01, SRV02, CL01.

TASKS

- 1 **Register an SPN on a service account. [DC01]** setspn -S checks for duplicates before adding.

```
New-ADUser svc-intranet -Path
'OU=ServiceAccounts,OU=Greenfield,DC=greenfieldcorp,DC=lab' -AccountPassword
(Read-Host -AsSecureString) -Enabled $true
setspn -S HTTP/intranet.greenfieldcorp.lab GREENFIELD\svc-intranet
setspn -L GREENFIELD\svc-intranet
```

- 2 **Create and detect a duplicate SPN.**

```
setspn -A HTTP/intranet.greenfieldcorp.lab GREENFIELD\gmsa-webapp$
setspn -X
setspn -D HTTP/intranet.greenfieldcorp.lab GREENFIELD\gmsa-webapp$
```

- 3 **Trace the ticket flow. [CL01 as maria.lopez]**

```
klist purge
klist                # empty
dir \\SRV01\c$ 2>$null; dir \\DC01\SYSTEM
klist                # krbtgt TGT + cifs/DC01 service ticket
klist tgt
```


- 4 **Inspect encryption types.** Confirm AES256 (0x12) for both the TGT and service tickets.

```
klist | Select-String 'Server|KerbTicket Encryption'
```

- 5 **Configure constrained delegation (KCD).** [DC01] SRV01 may delegate to CIFS on SRV02 only, using protocol transition.

```
Set-ADComputer SRV01 -Add @{'msDS-AllowedToDelegateTo'=@('cifs/SRV02.greenfieldcorp.lab','cifs/SRV02')}
Set-ADAccountControl SRV01 -TrustedToAuthForDelegation $true
Get-ADComputer SRV01 -Properties msDS-AllowedToDelegateTo,
TrustedToAuthForDelegation
```

- 6 **Replace it with resource-based constrained delegation.** RBCD is configured on the resource (SRV02), not the front end, and does not need Domain Admin.

```
Set-ADComputer SRV01 -Clear msDS-AllowedToDelegateTo; Set-ADAccountControl SRV01 -TrustedToAuthForDelegation $false
Set-ADComputer SRV02 -PrincipalsAllowedToDelegateToAccount (Get-ADComputer SRV01)
Get-ADComputer SRV02 -Properties PrincipalsAllowedToDelegateToAccount
```

- 7 **Test the double hop.** [CL01] From a session on SRV01, access a share on SRV02. Without delegation this fails with Access denied.

```
Invoke-Command -ComputerName SRV01 -ScriptBlock { Get-ChildItem \\SRV02\c$ |
Select-Object -First 3 }
```

- 8 **Find unconstrained delegation.** Only DCs should appear.

```
Get-ADComputer -Filter 'TrustedForDelegation -eq $true' | Select-Object Name
Get-ADUser -Filter 'TrustedForDelegation -eq $true' | Select-Object Name
```

- 9 **Audit NTLM.** [DC01] Enable via GPO DC-C-NTLMAudit-v1: Security Options > Network security: Restrict NTLM: Audit NTLM authentication in this domain = Enable all; Audit incoming NTLM traffic = Enable auditing for all accounts.

```
Get-WinEvent -LogName 'Microsoft-Windows-NTLM/Operational' -MaxEvents 20 | Format-Table TimeCreated, Id, Message -Wrap
Get-WinEvent -FilterHashtable @{'LogName'='Security';Id=4624} -MaxEvents 200 |
Where-Object { $_.Properties[10].Value -eq 'NTLM' } | Measure-Object
```

EXPECTED OUTPUT

```
#0> Client: maria.lopez @ GREENFIELDCORP.LAB
      Server: krbtgt/GREENFIELDCORP.LAB @ GREENFIELDCORP.LAB
      KerbTicket Encryption Type: AES-256-CTS-HMAC-SHA1-96
#1> Server: cifs/DC01 @ GREENFIELDCORP.LAB
setspn -X: found 1 group of duplicate SPNs (then 0 after removal)
PrincipalsAllowedToDelegateToAccount : {CN=SRV01,OU=Servers,...}
```

TROUBLESHOOTING

KRB_AP_ERR_MODIFIED (event 4)	Duplicate or wrong SPN: the ticket was encrypted for a different account. Run setspn -X and fix.
KRB_AP_ERR_SKEW	Clock skew > 5 minutes. Check w32tm /query /status on client and PDC.
Double hop still fails after RBCD	The client has a cached ticket from before. klist purge on CL01 and klist -li 0x3e7 purge on SRV01.

KNOWLEDGE CHECK

Q1 Walk through AS-REQ, TGS-REQ, and AP-REQ in your own words.

Q2 Why is unconstrained delegation dangerous?

Q3 Name two 2025 improvements that reduce legacy Netlogon/NTLM dependence.

SNAPSHOT ▸ L17-END-kerberos (DC01, SRV01, SRV02, CL01)

Security and Hardening

Securing the Directory

Apply the security baseline, enforce LDAP signing, and verify with a test bind.

OBJECTIVES

- Import the Microsoft Security Baseline for Windows Server 2025
- Enforce LDAP signing and channel binding
- Verify SMB signing and encryption settings
- Test simple and signed binds with ldp.exe

PREREQUISITES

L17-END-kerberos. Download the Microsoft Security Compliance Toolkit (Windows Server 2025 baseline + LGPO.exe) to C:\SCT on DC01.

TASKS

1 Inventory privileged groups. [DC01]

```
'Enterprise Admins','Schema Admins','Domain Admins','Administrators','Account Operators','Backup Operators','Server Operators','Print Operators' |  
  ForEach-Object { [pscustomobject]@{ Group=$_; Members=(Get-ADGroupMember $_ -  
    Recursive).SamAccountName -join ',' } }
```

2 Empty Schema Admins and clean operator groups. Membership should be empty except for break-glass accounts, added only when needed.

```
Get-ADGroupMember 'Schema Admins' | ForEach-Object { Remove-ADGroupMember 'Schema Admins' $_ -Confirm:$false }
```

3 Import the baseline GPOs. The baseline ships a Baseline-ADImport.ps1 script that creates the GPOs unlinked.

```
Set-Location 'C:\SCT\Windows Server 2025 Security Baseline\Scripts'  
.\Baseline-ADImport.ps1  
Get-GPO -All | Where-Object DisplayName -like 'MSFT*' | Select-Object DisplayName
```

4 Link baseline GPOs. DC baseline to Domain Controllers OU; member server baseline to OU=Servers. Review Policy Analyzer output before linking in production.

```
New-GPLink -Name 'MSFT Windows Server 2025 - Domain Controller' -Target 'OU=Domain
Controllers,DC=greenfieldcorp,DC=lab'
New-GPLink -Name 'MSFT Windows Server 2025 - Member Server' -Target
'OU=Servers,OU=Computers,OU=Greenfield,DC=greenfieldcorp,DC=lab'
```

- 5 **Enforce LDAP signing and channel binding.** Default Domain Controllers Policy > Security Options: "Domain controller: LDAP server signing requirements" = Require signing; "LDAP server channel binding token requirements" = Always. Or via registry for testing:

```
Set-ItemProperty HKLM:\SYSTEM\CurrentControlSet\Services\NTDS\Parameters -Name
LDAPServerIntegrity -Value 2 -Type DWord
Set-ItemProperty HKLM:\SYSTEM\CurrentControlSet\Services\NTDS\Parameters -Name
LdapEnforceChannelBinding -Value 2 -Type DWord
Invoke-Command DC02 { gpupdate /force }
```

- 6 **Test an unsigned simple bind.** [CL01] ldp.exe > Connection > Connect DC01, port 389 > Bind > Simple bind as maria.lopez. It must fail with error 8 (Strong auth required).
- 7 **Test a signed bind.** ldp.exe > Bind > "Bind as currently logged on user" with Sign checked. It must succeed.
- 8 **Check for unsigned-bind clients.** [DC01] Event 2889 lists clients that attempted unsigned binds (enable diagnostics first).

```
Set-ItemProperty HKLM:\SYSTEM\CurrentControlSet\Services\NTDS\Diagnostics -Name
'16 LDAP Interface Events' -Value 2
Get-WinEvent -FilterHashtable @{{LogName='Directory Service';Id=2887,2889} -
MaxEvents 20
```

- 9 **Verify SMB hardening.**

```
Get-SmbServerConfiguration | Select-Object RequireSecuritySignature, EncryptData,
RejectUnencryptedAccess, EnableSMB1Protocol
Get-SmbClientConfiguration | Select-Object RequireSecuritySignature, BlockNTLM
```

- 10 **Disable legacy ciphers and SMB1 checks.**

```
Get-WindowsFeature FS-SMB1 | Select-Object Name, InstallState
Get-ADDomainController -Filter * | ForEach-Object { Get-ADComputer $_.Name -
Properties msDS-SupportedEncryptionTypes } | Select-Object Name, msDS-
SupportedEncryptionTypes
```

EXPECTED OUTPUT

```
ldp simple bind: res = ldap_simple_bind_s(...) Error <8>: Strong Authentication
Required
ldp signed bind: Authenticated as: 'GREENFIELD\maria.lopez'.
RequireSecuritySignature : True   EnableSMB1Protocol : False
FS-SMB1 Available (not installed)
```

TROUBLESHOOTING

Applications break after enforcing LDAP signing	Roll back to value 1, collect event 2889 for 48 hours, fix each client, then enforce again.
Baseline-ADImport.ps1 not found	Extract the zip; the Scripts folder is inside the Windows Server 2025 baseline folder.
Cannot RDP after baseline	The baseline restricts "Allow log on through RDS". Add GG-T0-Admins to the DC policy or use the VirtualBox console.

KNOWLEDGE CHECK

- Q1 Why is compromise of any single DC equal to compromise of the whole forest?
- Q2 What does LDAP channel binding protect against that signing alone does not?
- Q3 Which attributes does 2025 require be read only over encrypted LDAP?

SNAPSHOT ▸ L18-END-baseline (DC01, DC02, CL01)

LAB 19

LESSON 19

Understanding Active Directory Attacks

In the isolated lab, observe a Kerberoasting attempt in the logs and identify the artifacts.

OBJECTIVES

- Enable Kerberos service ticket auditing
- Generate the ticket-request pattern Kerberoasting relies on, using built-in .NET only
- Identify the 4769 artifacts that distinguish it from normal traffic
- Map an attack path from a user to Domain Admin on paper

PREREQUISITES

ISOLATED LAB ONLY. L18-END-baseline. No offensive tooling is used; no password cracking is performed.

TASKS

1 Enable auditing. [DC01]

```
auditpol /set /subcategory:"Kerberos Service Ticket Operations" /success:enable  
/failure:enable  
auditpol /set /subcategory:"Kerberos Authentication Service" /success:enable  
/failure:enable  
auditpol /get /category:"Account Logon"
```

- 2 **Create a deliberately weak rostable account.** RC4 allowed and an SPN: the classic vulnerable pattern.

```
New-ADUser svc-olddb -Path  
'OU=ServiceAccounts,OU=Greenfield,DC=greenfieldcorp,DC=lab' -AccountPassword  
(Read-Host -AsSecureString) -Enabled $true -PasswordNeverExpires $true  
Set-ADUser svc-olddb -ServicePrincipalNames  
@{'Add='MSSQLSvc/sql01.greenfieldcorp.lab:1433'} -KerberosEncryptionType RC4,AES256
```

- 3 **Enumerate SPN accounts as a normal user. [CL01 as maria.lopez]** This is the reconnaissance step: any authenticated user can read SPNs.

```
([adsisearcher]'(&(objectCategory=user)(servicePrincipalName=*))').FindAll() |  
ForEach-Object { $_.Properties.samaccountname; $_.Properties.serviceprincipalname  
}
```

- 4 **Request service tickets for those SPNs.** Requesting a ticket is legitimate Kerberos behavior; attackers do it in bulk and then crack offline. We stop at the request.

```
Add-Type -AssemblyName System.IdentityModel  
'MSSQLSvc/sql01.greenfieldcorp.lab:1433','HTTP/intranet.greenfieldcorp.lab' |  
ForEach-Object {  
    New-Object System.IdentityModel.Tokens.KerberosRequestorSecurityToken -  
    ArgumentList $_ }  
klist | Select-String 'Server|Encryption'
```

- 5 **Find the artifacts. [DC01]** Look for 4769 with Ticket Encryption Type 0x17 (RC4) for user-owned SPNs, many SPNs from one user in a short window.

```
Get-WinEvent -FilterHashtable @{LogName='Security';Id=4769;StartTime=(Get-Date).AddMinutes(-30)} |
  ForEach-Object { $x=[xml]$_ .ToXml(); [pscustomobject]@{
    Time=$_ .TimeCreated
    User=($x.Event.EventData.Data | Where-Object Name -eq
'TargetUserName').'#text'
    Service=($x.Event.EventData.Data | Where-Object Name -eq
'ServiceName').'#text'
    EncType=($x.Event.EventData.Data | Where-Object Name -eq
'TicketEncryptionType').'#text'
    Client=($x.Event.EventData.Data | Where-Object Name -eq 'IpAddress').'#text' }
  } |
  Where-Object Service -notlike '*$' | Format-Table -AutoSize
```

6 Check AS-REP roasting exposure. Accounts without pre-auth required should not exist.

```
Get-ADUser -Filter 'DoesNotRequirePreAuth -eq $true' | Select-Object
SamAccountName
```

7 Check DCSync rights. Only DCs and Domain/Enterprise Admins should hold Replicating Directory Changes All.

```
(Get-Acl 'AD:DC=greenfieldcorp,DC=lab').Access |
  Where-Object { $_.ObjectType -in '1131f6ad-9c07-11d1-f79f-
00c04fc2dcd2', '1131f6aa-9c07-11d1-f79f-00c04fc2dcd2' } |
  Select-Object IdentityReference, ObjectType
```

8 Check the BadSuccessor exposure (dMSA create rights). Any non-Tier-0 principal with CreateChild on an OU can create a dMSA and set its predecessor link. List who can create child objects on OUs.

```
Get-ADOrganizationalUnit -Filter * | ForEach-Object {
  $dn=$_ .DistinguishedName
  (Get-Acl "AD:$dn").Access | Where-Object { $_.ActiveDirectoryRights -match
'CreateChild|GenericAll|WriteDacl' -and $_.IdentityReference -notmatch 'Domain
Admins|Enterprise Admins|SYSTEM|Administrators' } |
  Select-Object @{n='OU';e={$dn}}, IdentityReference, ActiveDirectoryRights }
```

9 Remediate svc-oldsql.

```
Set-ADUser svc-oldsql -KerberosEncryptionType AES256
Set-ADUser svc-oldsql -ServicePrincipalNames
@{Remove='MSSQLSvc/sql01.greenfieldcorp.lab:1433'}
Disable-ADAccount svc-oldsql
```

- 10 Draw the attack graph.** Nodes: maria.lopez → (reads SPN) svc-oldsq1 → (weak password) → SQL host local admin → cached Tier 0 creds → Domain Admins. Mark each edge with the control that breaks it.

EXPECTED OUTPUT

```
Time           User           Service      EncType Client
14:02:11      maria.lopez   svc-oldsq1   0x17    ::ffff:10.10.10.50
14:02:11      maria.lopez   svc-intranet 0x12    ::ffff:10.10.10.50
DoesNotRequirePreAuth: (no results)
```

TROUBLESHOOTING

No 4769 events Audit policy overridden by the baseline GPO. Set Advanced Audit Policy in the DC GPO, then gpupdate /force.

Ticket shows AES even for svc-oldsq1 The domain or client disallows RC4. That is the desired hardened state; note it and move on.

KNOWLEDGE CHECK

- Q1 Why does the 0x17 encryption type make a ticket easier to crack offline?
- Q2 Explain in defensive terms why BadSuccessor needed only OU CreateChild rights.
- Q3 Why do attackers think in graphs rather than lists?

SNAPSHOT ▸ L19-END-attack-observed (DC01, CL01)

LAB 20

LESSON 20

Detecting and Defending Against AD Attacks

Configure the SACLs and audit policy to detect a dMSA-creation event, then trigger and catch it.

OBJECTIVES

- Enable advanced audit subcategories for directory changes
- Set SACLs for dMSA creation and msDS-ManagedAccountPrecededByLink writes
- Trigger and capture events 5137 and 5136
- Restrict dMSA creation to Tier 0

PREREQUISITES

L19-END-attack-observed. DC01, DC02.

TASKS

- 1 Enable directory change auditing. [DC01]** Set in the DC GPO for persistence: Advanced Audit Policy > DS Access.


```
auditpol /set /subcategory:"Directory Service Changes" /success:enable
auditpol /set /subcategory:"Directory Service Access" /success:enable
/failure:enable
auditpol /set /subcategory:"Security Group Management" /success:enable
auditpol /get /category:"DS Access"
```

- 2 **Add a SACL for dMSA object creation at the domain root.** Audits Everyone creating msDS-DelegatedManagedServiceAccount objects anywhere in the domain.

```
$root='AD:DC=greenfieldcorp,DC=lab'
$acl=Get-Acl $root -Audit
$dmsaClass=[guid](Get-ADObject -SearchBase (Get-ADRootDSE).schemaNamingContext -
LDAPFilter '(LDAPDisplayName=msDS-DelegatedManagedServiceAccount)' -Properties
schemaIDGUID).schemaIDGUID
$rule=New-Object System.DirectoryServices.ActiveDirectoryAuditRule(
[System.Security.Principal.SecurityIdentifier]'S-1-1-
0','CreateChild','Success',$dmsaClass,'All')
$acl.AddAuditRule($rule); Set-Acl $root $acl
```

- 3 **Add a SACL for writes to the predecessor link.**

```
$attr=(Get-ADObject -SearchBase (Get-ADRootDSE).schemaNamingContext -LDAPFilter
'(LDAPDisplayName=msDS-ManagedAccountPrecededByLink)' -Properties
schemaIDGUID).schemaIDGUID
$rule2=New-Object System.DirectoryServices.ActiveDirectoryAuditRule(
[System.Security.Principal.SecurityIdentifier]'S-1-1-
0','WriteProperty','Success',[guid]$attr,'All')
$acl=Get-Acl $root -Audit; $acl.AddAuditRule($rule2); Set-Acl $root $acl
(Get-Acl $root -Audit).Audit | Select-Object IdentityReference,
ActiveDirectoryRights, ObjectType
```

- 4 **Trigger: create a test dMSA and set its link.**

```
New-ADServiceAccount dmsa-canary -DNSHostName dmsa-canary.greenfieldcorp.lab -
CreateDelegatedServiceAccount -Path
'OU=ServiceAccounts,OU=Greenfield,DC=greenfieldcorp,DC=lab'
Set-ADServiceAccount dmsa-canary -Replace @{msDS-ManagedAccountPrecededByLink'=
(Get-ADUser svc-olddb).DistinguishedName}
```

- 5 **Catch the events.**

```
Get-WinEvent -FilterHashtable @{LogName='Security';Id=5137,5136;StartTime=(Get-Date).AddMinutes(-10)} |
    Select-Object TimeCreated, Id, @{n='Summary';e={$_.Message -split
[Environment]::NewLine | Select-String 'DN:|LDAP Display Name:|Class:|Account
Name:') -join ' ' | '}} | Format-List
Get-WinEvent -FilterHashtable @{LogName='Directory Service';Id=2946} -MaxEvents 5
-ErrorAction SilentlyContinue
```

- 6 **Restrict who can create dMSAs.** Deny CreateChild for the dMSA class to Authenticated Users on each non-Tier-0 OU, and remove broad CreateChild ACEs found in Lab 19.

```
$deny=New-Object System.DirectoryServices.ActiveDirectoryAccessRule(
[System.Security.Principal.SecurityIdentifier]'S-1-5-
11','CreateChild','Deny',$dmsaClass,'All')
Get-ADOrganizationalUnit -SearchBase 'OU=Greenfield,DC=greenfieldcorp,DC=lab' -
Filter * | Where-Object DistinguishedName -notlike '*OU=Tier0*' | ForEach-Object {
    $p="AD:$($_.DistinguishedName)"; $a=Get-Acl $p; $a.AddAccessRule($deny); Set-Acl
    $p $a }
```

- 7 **Detect DCSync and privileged group changes.** 4662 with the replication GUIDs from a non-DC account indicates DCSync; 4728/4732/4756 indicate group adds.

```
Get-WinEvent -FilterHashtable @{LogName='Security';Id=4662} -MaxEvents 500 |
    Where-Object { $_.Message -match '1131f6ad-9c07-11d1-f79f-00c04fc2dcd2' -and
    $_.Message -notmatch 'DC0\d\$' }
Get-WinEvent -FilterHashtable @{LogName='Security';Id=4728,4732,4756} -MaxEvents
20 | Format-Table TimeCreated, Id -AutoSize
```

- 8 **Create a honey token.** An enabled account that no one uses, with an SPN. Any 4769 for it is an alert.

```
New-ADUser svc-backup-legacy -Description 'DO NOT USE' -Path
'OU=ServiceAccounts,OU=Greenfield,DC=greenfieldcorp,DC=lab' -AccountPassword
(ConvertTo-SecureString ([guid]::NewGuid().ToString()+'Aa1!') -AsPlainText -Force)
-Enabled $true
Set-ADUser svc-backup-legacy -ServicePrincipalNames
@{Add='MSSQLSvc/bkp01.greenfieldcorp.lab:1433'}
```

- 9 **Clean up the canary.**

```
Remove-ADServiceAccount dmsa-canary -Confirm:$false
```

EXPECTED OUTPUT

```
Id : 5137 A directory service object was created. Class: msDS-DelegatedManagedServiceAccount DN: CN=dmsa-canary,OU=ServiceAccounts,...
Id : 5136 A directory service object was modified. LDAP Display Name: msDS-ManagedAccountPrecededByLink
```

TROUBLESHOOTING

No 5137/5136 events

Either the subcategory is not enabled (auditpol /get) or the SACL did not save. Re-read with Get-Acl -Audit.

Set-Acl: This security ID may not be assigned as the owner

Run from an elevated session as a Domain Admin on a DC.

Too many 4662 events

Filter on the replication GUIDs and forward only those to the SIEM.

KNOWLEDGE CHECK

- Q1 Why is a SACL on msDS-ManagedAccountPrecededByLink more targeted than auditing all dMSA writes?
- Q2 What is the difference between Directory Service Access (4662) and Directory Service Changes (5136)?
- Q3 Why does a honey token produce almost no false positives?

SNAPSHOT ▸ L20-END-detection (DC01, DC02)

LAB 21

LESSON 21

Certificate Services and PKI Basics

Stand up a two-tier CA in the lab and issue an LDAPS certificate to your DC.

OBJECTIVES

- Build an offline standalone root CA
- Build an enterprise issuing CA on SRV01
- Publish CRLs and AIA to an HTTP location
- Issue a DC certificate and verify LDAPS on 636

PREREQUISITES

L20-END-detection. New VM ROOTCA (Server Core, workgroup, 10.10.10.30, never domain-joined). SRV01.

TASKS

1 Prepare CAPolicy.inf on ROOTCA. [ROOTCA]

```
@'
[Version]
Signature="$Windows NT$"
[Certsrv_Server]
RenewalKeyLength=4096
RenewalValidityPeriod=Years
RenewalValidityPeriodUnits=10
CRLPeriod=Months
CRLPeriodUnits=6
CRLDeltaPeriodUnits=0
LoadDefaultTemplates=0
'@ | Set-Content C:\Windows\CAPolicy.inf
```

2 Install the root CA.

```
Install-WindowsFeature ADCS-Cert-Authority
Install-AdcsCertificationAuthority -CAType StandaloneRootCA -CACommonName
'Greenfield Root CA' `
    -KeyLength 4096 -HashAlgorithmName SHA256 -ValidityPeriod Years -
ValidityPeriodUnits 10 -Force
certutil -setreg CA\DSConfigDN "CN=Configuration,DC=greenfieldcorp,DC=lab"
certutil -setreg CA\ValidityPeriodUnits 5
certutil -setreg CA\CRLPublicationURLs
"1:C:\Windows\System32\CertSrv\CertEnroll\%3%8%9.crl\n2:http://pki.greenfieldcorp.
lab/CertEnroll/%3%8%9.crl"
certutil -setreg CA\CACertPublicationURLs
"1:C:\Windows\System32\CertSrv\CertEnroll\%1_%3%4.crt\n2:http://pki.greenfieldcorp
.lab/CertEnroll/%1_%3%4.crt"
Restart-Service certsvc; certutil -crl
```

3 Prepare the HTTP publication point. [SRV01, DC01]

```
Install-WindowsFeature Web-Server
New-Item C:\inetpub\wwwroot\CertEnroll -ItemType Directory
# DC01:
Add-DnsServerResourceRecordCName -ZoneName greenfieldcorp.lab -Name pki -
HostNameAlias SRV01.greenfieldcorp.lab
```

4 Publish root cert and CRL to AD and HTTP. Copy

C:\Windows\System32\CertSrv\CertEnroll* from ROOTCA to SRV01 (use a virtual USB/shared folder, then disconnect).

```
Copy-Item <media>\*.cr* C:\inetpub\wwwroot\CertEnroll\
certutil -f -dspublish "C:\inetpub\wwwroot\CertEnroll\ROOTCA_Greenfield Root
CA.crt" RootCA
certutil -f -dspublish "C:\inetpub\wwwroot\CertEnroll\Greenfield Root CA.crl"
ROOTCA
```

5 Install the enterprise issuing CA. [SRV01]

```
Install-WindowsFeature ADCS-Cert-Authority -IncludeManagementTools
Install-AdcsCertificationAuthority -CAType EnterpriseSubordinateCA -CACommonName
'Greenfield Issuing CA 01' `
    -KeyLength 2048 -HashAlgorithmName SHA256 -OutputCertRequestFile C:\issuing.req
-Force
```

6 Sign the request on the root and install it.

```
# ROOTCA:
certreq -submit C:\issuing.req          # note RequestId
certutil -resubmit <RequestId>
certreq -retrieve <RequestId> C:\issuing.cer
# SRV01:
certutil -installcert C:\issuing.cer
Start-Service certsvc
certutil -setreg CA\CRLPublicationURLs
"65:C:\Windows\System32\CertSrv\CertEnroll\%3%8%9.crl\n6:http://pki.greenfieldcorp
.lab/CertEnroll/%3%8%9.crl\n65:C:\inetpub\wwwroot\CertEnroll\%3%8%9.crl"
Restart-Service certsvc; certutil -crl
```

7 Shut down ROOTCA. Power off and take the snapshot. It is only powered on to renew the issuing CA or publish a new CRL (every 6 months).

8 Enable autoenrollment. [DC01] GPO DC-C-Autoenroll-v1 on OU=Domain Controllers: Computer > Policies > Windows Settings > Security Settings > Public Key Policies > Certificate Services Client - Auto-Enrollment = Enabled, renew + update. The default Kerberos Authentication template is published on the issuing CA.

```
certutil -CATemplates -config "SRV01.greenfieldcorp.lab\Greenfield Issuing CA 01"
certutil -pulse; gpupdate /force
```

9 Verify the certificate and LDAPS.

```
Get-ChildItem Cert:\LocalMachine\My | Select-Object Subject, Issuer, NotAfter,
@{n='EKU';e={$_.EnhancedKeyUsageList.FriendlyName -join ', '}}
Test-NetConnection DC01 -Port 636
certutil -verify -urlfetch (Get-ChildItem Cert:\LocalMachine\My | Select-Object -
First 1).PSPath
```

- 10 Audit template permissions.** No template should grant Enroll to Domain Users together with "Supply in the request" and a Client Authentication EKU.

```
certutil -v -dstemplate | Select-String 'CN=|msPKI-Certificate-Name-Flag|pKIExtendedKeyUsage'
```

EXPECTED OUTPUT

```
Subject : CN=DC01.greenfieldcorp.lab
Issuer  : CN=Greenfield Issuing CA 01, DC=greenfieldcorp, DC=lab
EKU     : KDC Authentication,Smart Card Logon,Server Authentication,Client Authentication
TcpTestSucceeded : True (port 636)
certutil -verify: Leaf certificate revocation check passed
```

TROUBLESHOOTING

Issuing CA will not start: revocation server offline	The root CRL is not reachable at the HTTP path. Check the pki CNAME, IIS, and that the .crl file name matches.
IIS returns 404.11 for delta CRL (+ in name)	Set allowDoubleEscaping: Set-WebConfigurationProperty -Filter system.webServer/security/requestFiltering -Name allowDoubleEscaping -Value true.
DC did not get a certificate	Template not published or DC lacks Autoenroll permission. Run certutil -pulse and check Application log source CertificateServicesClient-AutoEnrollment.

KNOWLEDGE CHECK

- Q1 Why is the root CA kept offline?
- Q2 What are CDP and AIA used for during chain validation?
- Q3 Describe the template settings combination that creates an ESC1-style risk.

SNAPSHOT ▸ L21-END-pki (ROOTCA, SRV01, DC01)

Windows LAPS and Local Account Security

Deploy LAPS to member servers, confirm rotation, and audit a password retrieval.

OBJECTIVES

- Extend the schema and set permissions for Windows LAPS
- Configure LAPS by GPO with encryption
- Force and verify rotation
- Audit password retrieval

PREREQUISITES

L21-END-pki. Windows LAPS is built in; the legacy LAPS MSI is not needed. Domain functional level 2016+ is required for password encryption.

TASKS

1 Extend the schema. [DC01]

```
Update-LapsADSchema -Verbose
Get-ADObject -SearchBase (Get-ADRootDSE).schemaNamingContext -Filter "name -like 'ms-LAPS-*'" | Select-Object Name
```

2 Grant computers self-write on the Servers OU.

```
$srv='OU=Servers,OU=Computers,OU=Greenfield,DC=greenfieldcorp,DC=lab'
Set-LapsADComputerSelfPermission -Identity $srv
```

3 Create the reader group and grant read.

```
New-ADGroup GG-T1-LAPSReaders -GroupScope Global -Path
'OU=Tier1,OU=Admin,OU=Greenfield,DC=greenfieldcorp,DC=lab'
Set-LapsADReadPasswordPermission -Identity $srv -AllowedPrincipals 'GREENFIELD\GG-T1-LAPSReaders'
Find-LapsADExtendedRights -Identity $srv
```

4 Create the LAPS GPO. SRV-C-LAPS-v1 linked to OU=Servers. Computer > Policies > Administrative Templates > System > LAPS: Configure password backup directory = Active Directory; Password Settings = complexity Large+small+numbers+specials, length 20, age 30; Enable password encryption = Enabled, authorized decryptors = GREENFIELD\GG-T1-LAPSReaders; Post-authentication actions = Reset password and logoff, 8 hours.

```
New-GPO SRV-C-LAPS-v1 | New-GPLink -Target $srv
$k='HKLM\Software\Microsoft\Windows\CurrentVersion\Policies\LAPS'
Set-GPRegistryValue -Name SRV-C-LAPS-v1 -Key $k -ValueName BackupDirectory -Type
DWord -Value 2
Set-GPRegistryValue -Name SRV-C-LAPS-v1 -Key $k -ValueName PasswordLength -Type
DWord -Value 20
Set-GPRegistryValue -Name SRV-C-LAPS-v1 -Key $k -ValueName PasswordAgeDays -Type
DWord -Value 30
Set-GPRegistryValue -Name SRV-C-LAPS-v1 -Key $k -ValueName
ADPasswordEncryptionEnabled -Type DWord -Value 1
Set-GPRegistryValue -Name SRV-C-LAPS-v1 -Key $k -ValueName
ADPasswordEncryptionPrincipal -Type String -Value 'GREENFIELD\GG-T1-LAPSReaders'
```

5 Apply and force rotation. [SRV01, SRV02]

```
gpupdate /force
Invoke-LapsPolicyProcessing
Reset-LapsPassword
Get-WinEvent -LogName 'Microsoft-Windows-LAPS/Operational' -MaxEvents 10 | Format-
Table TimeCreated, Id, Message -Wrap
```

6 Enable retrieval auditing. [DC01] Add a SACL for reading msLAPS-EncryptedPassword so each retrieval logs 4662.

```
$attr=(Get-ADObject -SearchBase (Get-ADRootDSE).schemaNamingContext -LDAPFilter
'(LDAPDisplayName=msLAPS-EncryptedPassword)' -Properties
schemaIDGUID).schemaIDGUID
$p="AD:$srv"; $a=Get-Acl $p -Audit
$a.AddAuditRule((New-Object
System.DirectoryServices.ActiveDirectoryAuditRule([System.Security.Principal.Secur
ityIdentifier]'S-1-1-0', 'ReadProperty', 'Success', [guid]$attr, 'Descendents'))
Set-Acl $p $a
```

7 Retrieve a password as an authorized reader.

```
Add-ADGroupMember GG-T1-LAPSReaders adm-t0-maria.lopez
# new logon as adm-t0-maria.lopez:
Get-LapsADPassword -Identity SRV02 -AsPlainText
```

8 Confirm an unauthorized user cannot read it.

```
# as maria.lopez on CL01:
Get-LapsADPassword -Identity SRV02 -AsPlainText
```

9 Find the audit event.


```
Get-WinEvent -FilterHashtable @{LogName='Security';Id=4662;StartTime=(Get-Date).AddMinutes(-15)} |  
Where-Object Message -match 'SRV02' | Select-Object -First 3 TimeCreated,  
@{n='Who';e={$_.Properties[1].Value}}
```

EXPECTED OUTPUT

```
ComputerName      : SRV02  
Account           : Administrator  
Password          : q8#Fv...(20 chars)  
PasswordUpdateTime : 9/23/2026 2:41:07 PM  
ExpirationTimestamp : 10/23/2026 2:41:07 PM  
Source            : EncryptedPassword  
DecryptionStatus   : Success  
(maria.lopez: DecryptionStatus : Unauthorized)
```

TROUBLESHOOTING

LAPS event 10059: policy says AD but schema not extended	Update-LapsADSchema did not replicate yet. repadmin /syncall /AdeP and retry.
Event 10017: failed to update AD, access denied	Set-LapsADComputerSelfPermission was run on the wrong OU.
DecryptionStatus: Unauthorized for the reader	Group membership needs a new logon token. Sign out and back in.

KNOWLEDGE CHECK

- Q1 How does a shared local Administrator password enable lateral movement?
- Q2 Why encrypt the LAPS password when AD ACLs already protect it?
- Q3 Where do LAPS reader groups belong in the tier model?

SNAPSHOT ▸ L22-END-laps (DC01, SRV01, SRV02)

Infrastructure Services and Lifecycle

LAB 23

LESSON 23

DHCP and IP Address Management

Configure a DHCP scope with failover between two servers and confirm dynamic DNS updates.

OBJECTIVES

- Install and authorize DHCP on SRV01 and SRV02
- Build a scope with exclusions, reservations, and options
- Configure load-balance failover
- Secure dynamic DNS updates with a dedicated credential

PREREQUISITES

L22-END-laps. CL01 will be switched from static to DHCP.

TASKS

1 Install DHCP. [SRV01, SRV02]

```
Install-WindowsFeature DHCP -IncludeManagementTools
netsh dhcp add securitygroups
Restart-Service dhcpserver
Set-ItemProperty HKLM:\SOFTWARE\Microsoft\ServerManager\Roles\12 -Name
ConfigurationState -Value 2
```

2 Authorize both servers in AD. [DC01]

```
Add-DhcpServerInDC -DnsName SRV01.greenfieldcorp.lab -IPAddress 10.10.10.20
Add-DhcpServerInDC -DnsName SRV02.greenfieldcorp.lab -IPAddress 10.10.10.21
Get-DhcpServerInDC
```

3 Create the scope. [SRV01]

```
Add-DhcpServerv4Scope -Name 'HQ-Clients' -StartRange 10.10.10.100 -EndRange
10.10.10.199 -SubnetMask 255.255.255.0 -LeaseDuration 8.00:00:00 -State Active
Add-DhcpServerv4ExclusionRange -ScopeId 10.10.10.0 -StartRange 10.10.10.100 -
EndRange 10.10.10.109
Set-DhcpServerv4OptionValue -ScopeId 10.10.10.0 -Router 10.10.10.1 -DnsServer
10.10.10.10,10.10.10.11 -DnsDomain greenfieldcorp.lab
```

4 Create a reservation for CL01.

```
$mac = (Invoke-Command CL01 { (Get-NetAdapter | Select-Object -First 1).MacAddress
})
Add-DhcpServerv4Reservation -ScopeId 10.10.10.0 -IPAddress 10.10.10.150 -ClientId
$mac -Name CL01
```

- 5 **Secure DNS dynamic updates.** A dedicated low-privilege account prevents DHCP from owning DNS records as the server computer account.

```
New-ADUser svc-dhcpdns -Path
'OU=ServiceAccounts,OU=Greenfield,DC=greenfieldcorp,DC=lab' -AccountPassword
(Read-Host -AsSecureString) -Enabled $true -PasswordNeverExpires $true
# on SRV01 and SRV02:
Set-DhcpServerDnsCredential -Credential (Get-Credential GREENFIELD\svc-dhcpdns) -
ComputerName localhost
Set-DhcpServerv4DnsSetting -ScopeId 10.10.10.0 -DynamicUpdates Always -
DeleteDnsRROnLeaseExpiry $true -UpdateDnsRRForOlderClients $true
```

- 6 **Configure failover. [SRV01]**

```
Add-DhcpServerv4Failover -Name HQ-Failover -PartnerServer SRV02.greenfieldcorp.lab
-ScopeId 10.10.10.0 `
  -LoadBalancePercent 50 -MaxClientLeadTime 01:00:00 -SharedSecret (Read-Host
'Secret') -AutoStateTransition $true -StateSwitchInterval 00:30:00
Get-DhcpServerv4Failover
Invoke-DhcpServerv4FailoverReplication -Name HQ-Failover -Force
```

- 7 **Switch CL01 to DHCP. [CL01]**

```
$if='Ethernet'
Remove-NetIPAddress -InterfaceAlias $if -AddressFamily IPv4 -Confirm:$false
Remove-NetRoute -InterfaceAlias $if -Confirm:$false
Set-NetIPInterface -InterfaceAlias $if -Dhcp Enabled
Set-DnsClientServerAddress -InterfaceAlias $if -ResetServerAddresses
ipconfig /renew; ipconfig /all | Select-String 'IPv4|DHCP Server|DNS Servers'
```

- 8 **Verify DNS registration and failover. [DC01]** Stop DHCP on SRV01, renew on CL01, confirm SRV02 answers.

```
Get-DnsServerResourceRecord -ZoneName greenfieldcorp.lab -Name CL01
Invoke-Command SRV01 { Stop-Service dhcpserver }
Invoke-Command CL01 { ipconfig /release; ipconfig /renew; (ipconfig /all | Select-
String 'DHCP Server') }
Invoke-Command SRV01 { Start-Service dhcpserver }
```

- 9 **Review logs and detect rogues.**

```
Get-Content C:\Windows\System32\dhcp\DhcpSrvLog-$(Get-Date).DayOfWeek.ToString().Substring(0,3)).log -Tail 20
Get-DhcpServerv4ScopeStatistics
```

EXPECTED OUTPUT

```
IPv4 Address. . . : 10.10.10.150(Preferred)
DHCP Server . . . : 10.10.10.20 (10.10.10.21 while SRV01 stopped)
HostName RecordType RecordData
CL01      A          10.10.10.150
State : Normal Mode : LoadBalance LoadBalancePercent : 50
```

TROUBLESHOOTING

Scope does not lease: server not authorized	Get-DhcpServerInDC must list the server. Re-run Add-DhcpServerInDC as Enterprise Admin, restart dhcpserver.
Failover state CommunicationInterrupted	TCP 647 blocked or clocks differ by more than 60 seconds. Check w32tm on both and the DHCP firewall rules.
DNS A record owned by the wrong account	Credential was set after the record was created. Delete the record and renew the lease.

KNOWLEDGE CHECK

- Q1 Why does AD require DHCP servers to be authorized?
- Q2 Compare hot standby and load-balance failover.
- Q3 What problem does the DHCP DNS update credential solve?

SNAPSHOT ▸ L23-END-dhcp (DC01, SRV01, SRV02, CL01)

LAB 24

LESSON 24

File Services, Storage, and Permissions

Build a departmental share with correct layered permissions, ABE, and shadow copies.

OBJECTIVES

- Create an ReFS/NTFS data volume and choose appropriately
- Apply share and NTFS permissions through AGDLP groups
- Enable access-based enumeration, quotas, and file screens
- Enable shadow copies and test self-service restore

PREREQUISITES

L23-END-dhcp. SRV01 with D: (NTFS) from Lab 14. Add a 20 GB disk for E:.

TASKS

1 Create the folder structure. [SRV01]

```
New-Item D:\Shares\Engineering\Designs, D:\Shares\Engineering\Process,  
D:\Shares\Engineering\Public -ItemType Directory -Force
```

2 Set NTFS permissions via groups. Remove inheritance, keep admins and SYSTEM, grant Modify/Read to Domain Local groups only.

```
$p='D:\Shares\Engineering'  
icacls $p /inheritance:r  
icacls $p /grant "SYSTEM:(OI)(CI)F" "BUILTIN\Administrators:(OI)(CI)F"  
icacls $p /grant "GREENFIELD\DL-ENG-Share-Modify:(OI)(CI)M"  
icacls $p /grant "GREENFIELD\DL-ENG-Share-Read:(OI)(CI)RX"  
icacls $p
```

3 Create the share with ABE and SMB encryption. Share permission: Authenticated Users Full; NTFS does the real work (most restrictive wins).

```
New-SmbShare -Name Engineering -Path $p -FullAccess 'Authenticated Users' -  
FolderEnumerationMode AccessBased -EncryptData $true  
Get-SmbShare Engineering | Select-Object Name, FolderEnumerationMode, EncryptData  
Get-SmbShareAccess Engineering
```

4 Restrict a subfolder. Only Designers can see Designs; ABE hides it from others.

```
New-ADGroup DL-ENG-Designs-Modify -GroupScope DomainLocal -Path  
'OU=Resource,OU=Groups,OU=Greenfield,DC=greenfieldcorp,DC=lab'  
Add-ADGroupMember DL-ENG-Designs-Modify -Members GG-ENG-Designers  
icacls D:\Shares\Engineering\Designs /inheritance:r /grant "SYSTEM:(OI)(CI)F"  
"BUILTIN\Administrators:(OI)(CI)F" "GREENFIELD\DL-ENG-Designs-Modify:(OI)(CI)M"
```

5 Install FSRM, add quota and file screen.

```
Install-WindowsFeature FS-Resource-Manager -IncludeManagementTools  
New-FsrmQuota -Path D:\Shares\Engineering -Size 10GB -Description 'ENG 10 GB hard'  
New-FsrmFileScreen -Path D:\Shares\Engineering -Template 'Block Executable Files'  
-Active
```

6 Enable shadow copies.

```
vssadmin add shadowstorage /for=D: /on=D: /maxsize=10%
vssadmin create shadow /for=D:
$a = New-ScheduledTaskAction -Execute vssadmin.exe -Argument 'create shadow /for=D:'
$t = New-ScheduledTaskTrigger -Daily -At 7am, 12pm
Register-ScheduledTask ShadowCopy-D -Action $a -Trigger $t -User SYSTEM -RunLevel Highest
```

7 Create a DFS namespace.

```
Install-WindowsFeature FS-DFS-Namespaces -IncludeManagementTools
New-Item C:\DFSRoots\Data -ItemType Directory; New-SmbShare Data -Path C:\DFSRoots\Data -ReadAccess Everyone
New-DfsnRoot -Path \\greenfieldcorp.lab\Data -TargetPath \\SRV01\Data -Type DomainV2
New-DfsnFolder -Path \\greenfieldcorp.lab\Data\Engineering -TargetPath \\SRV01\Engineering
```

8 Test as users. [CL01] As maria.lopez (Designer): sees Designs, can write. As priya.nair (Process): Designs is hidden. Copy notepad.exe: blocked.

```
Get-ChildItem \\greenfieldcorp.lab\Data\Engineering
New-Item \\greenfieldcorp.lab\Data\Engineering\Public\test.txt -Value 'v1'
Set-Content \\greenfieldcorp.lab\Data\Engineering\Public\test.txt 'v2'
```

9 Restore from a previous version. After a shadow copy runs, right-click test.txt > Properties > Previous Versions > Restore. Or list them:

```
vssadmin list shadows /for=D:
```

10 Check effective access.

```
Get-SmbConnection | Select-Object ServerName, ShareName, Dialect, Encrypted
Get-SmbServerConfiguration | Select-Object EncryptData, RequireSecuritySignature
```

EXPECTED OUTPUT

```
Name          FolderEnumerationMode EncryptedData
Engineering AccessBased          True
priya.nair> Get-ChildItem ... -> Process, Public (Designs hidden)
Copy notepad.exe -> Access is denied (File screen)
Dialect 3.1.1 Encrypted True
```

TROUBLESHOOTING

Users see folders they cannot open	ABE is off. Set-SmbShare Engineering - FolderEnumerationMode AccessBased.
Designer gets Access Denied	Token is stale or nesting is wrong. Check Get-ADGroupMember DL-ENG-Designs-Modify -Recursive and re-login.
Legacy client cannot connect	Encryption and signing are required in 2025. Update the client; do not weaken the server for it.

KNOWLEDGE CHECK

- Q1 How do share and NTFS permissions combine?
- Q2 When would you pick ReFS over NTFS for a data volume?
- Q3 Why must users be placed in groups rather than on ACLs directly?

SNAPSHOT ▸ L24-END-file-services (DC01, SRV01, CL01)

LAB 25

LESSON 25

Hyper-V and Virtualization

Enable nested Hyper-V, create a VM, and configure Hyper-V Replica.

OBJECTIVES	• Enable nested virtualization in VirtualBox • Install Hyper-V and create an internal virtual switch • Create a Generation 2 VM with dynamic memory and a differencing disk • Configure Hyper-V Replica between two hosts
PREREQUISITES	L24-END-file-services. Host must have the Windows hypervisor OFF (Lab 01, step 4) for VirtualBox nested virtualization. Raise SRV01 to 6 GB and SRV02 to 4 GB RAM while powered off.

TASKS

1 Enable nested VT-x. [host, VMs powered off]

```
VBoxManage modifyvm SRV01 --nested-hw-virt on --memory 6144 --cpus 2
VBoxManage modifyvm SRV02 --nested-hw-virt on --memory 4096 --cpus 2
VBoxManage startvm SRV01; VBoxManage startvm SRV02
```

2 Confirm support and install Hyper-V. [SRV01, SRV02]

```
(Get-CimInstance Win32_Processor).VirtualizationFirmwareEnabled  
systeminfo | Select-String 'Hyper-V'  
Install-WindowsFeature Hyper-V -IncludeManagementTools -Restart
```

- 3 **Create an internal switch. [SRV01]** Internal keeps nested VMs off the lab network; use External only if you need them on GFM-NAT.

```
New-VMSwitch -Name 'vInternal' -SwitchType Internal  
Get-VMSwitch | Select-Object Name, SwitchType
```

- 4 **Create a parent disk and a differencing child.**

```
New-Item D:\HyperV\Disks -ItemType Directory -Force  
New-VHD -Path D:\HyperV\Disks\base.vhdx -SizeBytes 20GB -Dynamic  
New-VHD -Path D:\HyperV\Disks\app01.vhdx -ParentPath D:\HyperV\Disks\base.vhdx -  
Differencing  
Get-VHD D:\HyperV\Disks\app01.vhdx | Select-Object VhdType, ParentPath
```

- 5 **Create a Generation 2 VM.** No OS is required for the replica exercise.

```
New-VM -Name APP01 -Generation 2 -MemoryStartupBytes 512MB -VHDPATH  
D:\HyperV\Disks\app01.vhdx -SwitchName vInternal -Path D:\HyperV  
Set-VMemory APP01 -DynamicMemoryEnabled $true -MinimumBytes 256MB -MaximumBytes  
1GB  
Set-VMFirmware APP01 -EnableSecureBoot On  
Get-VMIntegrationService APP01 | Select-Object Name, Enabled
```

- 6 **Practice checkpoints.** Production checkpoints use VSS inside the guest; standard checkpoints capture memory state.

```
Set-VM APP01 -CheckpointType Production  
Checkpoint-VM APP01 -SnapshotName 'pre-change'  
Get-VMSnapshot APP01  
Remove-VMSnapshot APP01 -Name 'pre-change'
```

- 7 **Enable the Replica server. [SRV02]**

```
Set-VMReplicationServer -ReplicationEnabled $true -AllowedAuthenticationType  
Kerberos -ReplicationAllowedFromAnyServer $true -DefaultStorageLocation D:\Replica  
New-Item D:\Replica -ItemType Directory -Force  
Enable-NetFirewallRule -DisplayName 'Hyper-V Replica HTTP Listener (TCP-In)'
```

- 8 **Enable replication for APP01. [SRV01]**


```
Enable-VMReplication -VMName APP01 -ReplicaServerName SRV02.greenfieldcorp.lab -  
ReplicaServerPort 80 -AuthenticationType Kerberos -ReplicationFrequencySec 300  
Start-VMInitialReplication -VMName APP01  
Measure-VMReplication APP01
```

9 Test failover.

```
# SRV02:  
Start-VMFailover -VMName APP01 -AsTest  
Get-VM 'APP01 - Test' | Select-Object Name, State  
Stop-VMFailover -VMName APP01
```

10 Check DC virtualization safeguards. [DC01] VM-GenerationID lets a DC detect restore/clone and discard its RID pool and invocation ID, preventing USN rollback. VirtualBox does not expose VM-GenerationID; never restore a DC snapshot in production-like scenarios.

```
Get-ADDomainController -Identity DC01 | Select-Object Name, InvocationId  
Get-WinEvent -LogName 'Directory Service' -MaxEvents 200 | Where-Object Id -in  
2170,2168
```

EXPECTED OUTPUT

```
VhdType : Differencing ParentPath : D:\HyperV\Disks\base.vhdx  
Name State Health PrimaryServer ReplicaServer  
APP01 Replicating Normal SRV01.greenfieldcorp.lab SRV02.greenfieldcorp.lab  
APP01 - Test Off
```

TROUBLESHOOTING

Hyper-V install: processor does not have required virtualization capabilities	Host hypervisor still active or --nested-hw-virt not applied. Power off, re-apply, verify with systeminfo.
Enable-VMReplication: Kerberos authentication failed	Use FQDN for the replica server and confirm port 80 rule is enabled on SRV02.
Very slow nested performance	Expected. Keep nested VMs tiny and power them off after the lab.

KNOWLEDGE CHECK

- Q1 Compare fixed, dynamic, and differencing disks for performance and risk.
- Q2 What does VM-GenerationID protect against on a virtualized DC?
- Q3 Why are checkpoints not a backup?

Backup, Recovery, and Business Continuity

Delete an OU, recover it from the AD Recycle Bin, then perform an authoritative restore.

OBJECTIVES

- Take a system state backup of a DC
- Enable the AD Recycle Bin and recover deleted objects
- Perform a non-authoritative restore in DSRM followed by an authoritative restore of an OU
- Record tombstone lifetime and recovery documentation

PREREQUISITES

L25-END-hyperv. DC02 gets a second 40 GB disk for backups (VBoxManage createmedium + storageattach). DSRM password from Lab 11.

TASKS

1 Check tombstone lifetime. [DC01]

```
(Get-ADObject "CN=Directory Service,CN=Windows NT,CN=Services,$((Get-ADRootDSE).configurationNamingContext)" -Properties tombstoneLifetime).tombstoneLifetime
```

2 Install Windows Server Backup and back up system state. [DC02] Initialize the new disk as E:. Confirm your backup product supports the 32k page database format before Lab 27.

```
Install-WindowsFeature Windows-Server-Backup
Get-Disk | Where-Object PartitionStyle -eq RAW | Initialize-Disk -PassThru | New-Partition -DriveLetter E -UseMaximumSize | Format-Volume -FileSystem NTFS -NewFileSystemLabel Backup
wbadmin start systemstatebackup -backupTarget:E: -quiet
wbadmin get versions
```

3 Enable the Recycle Bin. [DC01] One-way change.

```
Enable-ADOptionalFeature 'Recycle Bin Feature' -Scope ForestOrConfigurationSet -Target greenfieldcorp.lab -Confirm:$false
Get-ADOptionalFeature 'Recycle Bin Feature' | Select-Object EnabledScopes
```

4 Delete the Sales OU with a test user.

```
New-ADUser dana.kim -Path
'OU=Sales,OU=Users,OU=Greenfield,DC=greenfieldcorp,DC=lab' -AccountPassword (Read-
Host -AsSecureString) -Enabled $true
Set-ADOrganizationalUnit
'OU=Sales,OU=Users,OU=Greenfield,DC=greenfieldcorp,DC=lab' -
ProtectedFromAccidentalDeletion $false
Remove-ADOrganizationalUnit
'OU=Sales,OU=Users,OU=Greenfield,DC=greenfieldcorp,DC=lab' -Recursive -
Confirm:$false
```

5 Recover from the Recycle Bin. Restore the parent OU first, then its children.

```
Get-ADObject -Filter 'isDeleted -eq $true -and Name -like "Sales*"' -
IncludeDeletedObjects | Restore-ADObject
Get-ADObject -Filter 'isDeleted -eq $true -and lastKnownParent -like "OU=Sales*"'
-IncludeDeletedObjects | Restore-ADObject
Get-ADUser dana.kim | Select-Object DistinguishedName, Enabled
```

6 Set up the authoritative restore scenario. Imagine the Recycle Bin were not enabled. Delete the OU again and let it replicate.

```
Set-ADOrganizationalUnit
'OU=Sales,OU=Users,OU=Greenfield,DC=greenfieldcorp,DC=lab' -
ProtectedFromAccidentalDeletion $false
Remove-ADOrganizationalUnit
'OU=Sales,OU=Users,OU=Greenfield,DC=greenfieldcorp,DC=lab' -Recursive -
Confirm:$false
repadmin /syncall /AdeP
```

7 Boot DC02 into DSRM. [DC02] Log on as .\Administrator with the DSRM password.

```
bcdedit /set safeboot dsrepair
Restart-Computer
```

8 Non-authoritative restore of system state.

```
wbadmin get versions
wbadmin start systemstaterecovery -version:<MM/DD/YYYY-HH:MM> -quiet
# do NOT reboot yet
```

9 Mark the OU authoritative. Increases version numbers so the restored objects win replication.

```
ntdsutil
activate instance ntds
authoritative restore
restore subtree "OU=Sales,OU=Users,OU=Greenfield,DC=greenfieldcorp,DC=lab"
quit
quit
```

10 Reboot normally and verify.

```
bcdedit /deletevalue safeboot
Restart-Computer
# after boot, on DC01:
repadmin /syncall /AdeP
Get-ADUser dana.kim
```

11 Restore back-links. ntdsutil writes an LDF for group memberships. Import it on DC02.

```
Get-ChildItem C:\Windows\System32\ar_*.ldf
ldifde -i -k -f C:\Windows\System32\ar_<timestamp>_links_greenfieldcorp.lab.ldf
```

EXPECTED OUTPUT

```
tombstoneLifetime : 180
EnabledScopes : {CN=Partitions,CN=Configuration,DC=greenfieldcorp,DC=lab, ...}
DistinguishedName :
CN=dana.kim,OU=Sales,OU=Users,OU=Greenfield,DC=greenfieldcorp,DC=lab
ntdsutil: Successfully updated 2 records.
```

TROUBLESHOOTING

wbadmin: backup target is on a critical volume	Backup target cannot be C:. Use the dedicated E: disk.
Restore-ADObject: parent does not exist	Restore the OU before its children, as in step 5.
DC02 boots normally instead of DSRM	bcdedit ran without elevation. Re-run elevated, or press F8 is not available; use shutdown /r /o > Troubleshoot.

KNOWLEDGE CHECK

- Q1 What is the difference between authoritative and non-authoritative restore?
- Q2 Why must you never restore a DC backup older than the tombstone lifetime?
- Q3 What does the Recycle Bin preserve that tombstone reanimation does not?

SNAPSHOT ▸ L26-END-recovery (DC01, DC02)

Upgrading and Migrating to Windows Server 2025

Add a 2025 DC to your domain, transfer roles, raise the functional level, and enable 32k pages.

OBJECTIVES

- Run pre-upgrade health checks
- Add DC03 and migrate FSMO roles
- Demote a DC cleanly
- Raise domain and forest functional levels to Windows Server 2025
- Enable the 32k database page optional feature

PREREQUISITES

L26-END-recovery. The forest is at Windows Server 2016 level (Lab 09). A verified system state backup that supports 32k pages. TPL-WS2025 for DC03.

TASKS

1 Pre-flight health. [DC01] Fix every error before proceeding.

```
dcdiag /e /q
repadmin /replsummary
repadmin /showbackup
Get-ADDomainController -Filter * | Select-Object Name, OperatingSystem, Site,
IsGlobalCatalog
```

2 Review schema version. Adprep runs automatically when the first 2025 DC is promoted into an older forest. Our forest is already 2025-schema (91).

```
Get-ADObject (Get-ADRootDSE).schemaNamingContext -Properties objectVersion
```

3 Build DC03. [host, DC03]

```
VBoxManage clonevm TPL-WS2025 --snapshot "L02-END-template-sysprepped" --options
link --name DC03 --register
VBoxManage modifyvm DC03 --memory 3072; VBoxManage startvm DC03
# DC03:
New-NetIPAddress -InterfaceAlias Ethernet -IPAddress 10.10.10.12 -PrefixLength 24
-DefaultGateway 10.10.10.1
Set-DnsClientServerAddress -InterfaceAlias Ethernet -ServerAddresses
10.10.10.10,10.10.10.11
Rename-Computer DC03 -Restart
```

4 Promote DC03.

```
Add-Computer -DomainName greenfieldcorp.lab -Credential GREENFIELD\Administrator -
Restart
Install-WindowsFeature AD-Domain-Services -IncludeManagementTools
Install-ADDSDomainController -DomainName greenfieldcorp.lab -InstallDns -SiteName
HQ-GRR -Credential (Get-Credential) -SafeModeAdministratorPassword (Read-Host -
AsSecureString) -Force
```

5 Transfer all roles to DC03.

```
Move-ADDirectoryServerOperationMasterRole -Identity DC03 -OperationMasterRole
0,1,2,3,4 -Confirm:$false
netdom query fsmo
```

6 Demote DC02 (simulating the old DC). Afterwards, rebuild DC02 as a fresh 2025 DC if you want three DCs; an in-place upgraded DC cannot use 32k pages.

```
# DC02:
Uninstall-ADDSDomainController -LocalAdministratorPassword (Read-Host -
AsSecureString) -DemoteOperationMasterRole -Confirm:$false
# DC01 after reboot:
Get-ADDomainController -Filter * | Select-Object Name
Get-DnsServerResourceRecord -ZoneName greenfieldcorp.lab | Where-Object {
$_.RecordData.IPv4Address -eq '10.10.10.11' }
```

7 Raise functional levels. One-way. All DCs must run 2025.

```
Set-ADDomainMode -Identity greenfieldcorp.lab -DomainMode Win2025Domain -
Confirm:$false
Set-ADForestMode -Identity greenfieldcorp.lab -ForestMode Win2025Forest -
Confirm:$false
(Get-ADDomain).DomainMode; (Get-ADForest).ForestMode
Get-ADObject (Get-ADRootDSE).defaultNamingContext -Properties msDS-Behavior-
Version
```

8 Back up before 32k. Irreversible. Take a fresh system state backup and VirtualBox snapshots of every DC now.

```
wbadmin start systemstatebackup -backupTarget:E: -quiet # on a DC with a backup
disk
VBoxManage snapshot DC01 take L27-PRE-32k; VBoxManage snapshot DC03 take L27-PRE-
32k
```

9 Enable 32k pages.

```
Get-ADOptionalFeature -Filter * | Select-Object Name, RequiredForestMode,
EnabledScopes
Enable-ADOptionalFeature 'Database 32k Pages Feature' -Scope
ForestOrConfigurationSet -Target greenfieldcorp.lab -Confirm:$false
Get-ADOptionalFeature 'Database 32k Pages Feature' | Select-Object EnabledScopes
```

- 10 Verify on each DC.** Each DC converts its database online; check the Directory Service log for page-size events and repair any object missing key attributes with fixupObjectState if reported.

```
Get-WinEvent -LogName 'Directory Service' -MaxEvents 50 | Where-Object Message -
match '32k|page size' | Format-Table TimeCreated, Id, Message -Wrap
dcdiag /e /q; repadmin /replsummary
```

EXPECTED OUTPUT

```
Schema master / ... / Infrastructure master : DC03.greenfieldcorp.lab
DomainMode : Windows2025Domain
ForestMode : Windows2025Forest
msDS-Behavior-Version : 10
Name : Database 32k Pages Feature EnabledScopes : {CN=Partitions,...}
```

TROUBLESHOOTING

Set-ADForestMode: not all DCs support the level	A demoted DC left metadata. Run ntdsutil > metadata cleanup or remove the server object from Sites and Services.
32k feature fails to enable	A DC was upgraded in place or is offline. Every DC must be a fresh 2025 promotion and reachable.
Clients still use DC02 IP	Update DHCP option 6 and static DNS settings to 10.10.10.10, 10.10.10.12.

KNOWLEDGE CHECK

- Q1 Why is migration usually safer than in-place upgrade for DCs?
- Q2 Why can an in-place upgraded DC not use 32k pages?
- Q3 What is your rollback plan after raising the forest functional level?

 SNAPSHOT ▶ L27-END-migrated-32k (DC01, DC02, DC03)

Automation, Monitoring, and Mastery

LAB 28

LESSON 28

Automating Administration with PowerShell

Build a scheduled reporting script that emails a weekly directory health summary.

OBJECTIVES

- Build a reusable module of reporting functions
 - Store credentials securely with the SecretManagement module
 - Run the report unattended as a gMSA scheduled task
 - Put scripts under Git source control
-

PREREQUISITES

L27-END-migrated-32k. SRV01 runs a lab SMTP relay (smtp4dev) on port 25; install from github.com/rnwood/smtp4dev and run as a service.

TASKS

1 Create the module folder. [SRV01]

```
$m="$env:ProgramFiles\WindowsPowerShell\Modules\GfmAdmin"  
New-Item $m -ItemType Directory -Force  
Install-WindowsFeature RSAT-AD-PowerShell
```

2 Write GfmAdmin.psm1.


```

function Get-GfmStaleAccount { param([int]$Days=90)
    Search-ADAccount -AccountInactive -TimeSpan ([timespan]::FromDays($Days)) -
UsersOnly |
    Where-Object Enabled | Select-Object Name, SamAccountName, LastLogonDate }
function Get-GfmPasswordAge { param([int]$Days=180)
    Get-ADUser -Filter 'Enabled -eq $true' -Properties PasswordLastSet,
PasswordNeverExpires |
    Where-Object { $_.PasswordNeverExpires -or $_.PasswordLastSet -lt (Get-
Date).AddDays(-$Days) } |
    Select-Object SamAccountName, PasswordLastSet, PasswordNeverExpires }
function Get-GfmPrivilegedMember {
    'Domain Admins','Enterprise Admins','Schema Admins','Administrators' | ForEach-
Object { $g=$_
    Get-ADGroupMember $g -Recursive | Select-Object @{n='Group';e={$g}},
SamAccountName } }
function Get-GfmReplicationHealth {
    Get-ADReplicationPartnerMetadata -Target * -Scope Domain -ErrorAction
SilentlyContinue |
    Select-Object Server, Partner, LastReplicationSuccess,
ConsecutiveReplicationFailures }
Export-ModuleMember -Function Get-Gfm*

```

3 Write the report script C:\Scripts\Send-WeeklyADReport.ps1.

```
[CmdletBinding(SupportsShouldProcess)]
param([string]$To='it-ops@greenfieldcorp.lab',
[string]$Smtip='SRV01.greenfieldcorp.lab',[string]$Log='C:\Scripts\Logs')
$ErrorActionPreference='Stop'
New-Item $Log -ItemType Directory -Force | Out-Null
Start-Transcript "$Log\report-$(Get-Date -Format yyyyMMdd).log"
try {
    Import-Module GfmAdmin
    $css='<style>body{font-family:Segoe UI}table{border-
collapse:collapse}td,th{border:1px solid #999;padding:4px}</style>'
    $body = $css +
        (Get-GfmReplicationHealth | ConvertTo-Html -Fragment -PreContent
'<h2>Replication</h2>') +
        (Get-GfmPrivilegedMember | ConvertTo-Html -Fragment -PreContent
'<h2>Privileged groups</h2>') +
        (Get-GfmStaleAccount | ConvertTo-Html -Fragment -PreContent '<h2>Stale
accounts (90d)</h2>') +
        (Get-GfmPasswordAge | ConvertTo-Html -Fragment -PreContent '<h2>Old /
non-expiring passwords</h2>')
    if ($PSCmdlet.ShouldProcess($To,'Send report')) {
        $msg = New-Object
System.Net.Mail.MailMessage('adreport@greenfieldcorp.lab',$To,"AD weekly health
$(Get-Date -Format d)",$body)
        $msg.IsBodyHtml = $true
        (New-Object System.Net.Mail.SmtpClient($Smtip,25)).Send($msg)
    }
} catch { Write-Error $_; exit 1 } finally { Stop-Transcript }
```

4 Dry-run, then send.

```
C:\Scripts\Send-WeeklyADReport.ps1 -WhatIf
C:\Scripts\Send-WeeklyADReport.ps1
Start-Process http://localhost:5000 # smtp4dev web UI
```

5 Create a gMSA for the task. [DC01]

```
New-ADServiceAccount gmsa-adreport -DNSHostName gmsa-adreport.greenfieldcorp.lab -
PrincipalsAllowedToRetrieveManagedPassword 'SRV01$' -Path
'OU=ServiceAccounts,OU=Greenfield,DC=greenfieldcorp,DC=lab'
# SRV01:
Install-ADServiceAccount gmsa-adreport; Test-ADServiceAccount gmsa-adreport
```

6 Schedule it as the gMSA. [SRV01] The gMSA needs "Log on as a batch job" (add via GPO on OU=Servers).

```
$a=New-ScheduledTaskAction -Execute powershell.exe -Argument '-NoProfile -
ExecutionPolicy RemoteSigned -File C:\Scripts\Send-WeeklyADReport.ps1'
$t=New-ScheduledTaskTrigger -Weekly -DaysOfWeek Monday -At 6am
$p=New-ScheduledTaskPrincipal -UserId 'GREENFIELD\gmsa-adreport$' -LogonType
Password
Register-ScheduledTask GFM-WeeklyADReport -Action $a -Trigger $t -Principal $p
Start-ScheduledTask GFM-WeeklyADReport; Get-ScheduledTaskInfo GFM-WeeklyADReport
```

- 7 Store other secrets securely.** Never put passwords in scripts. For secrets a gMSA cannot cover:

```
Install-Module Microsoft.PowerShell.SecretManagement,
Microsoft.PowerShell.SecretStore -Scope CurrentUser
Register-SecretVault -Name LabVault -ModuleName Microsoft.PowerShell.SecretStore -
DefaultVault
Set-Secret -Name SmtplRelay -Secret (Get-Credential)
Get-Secret SmtplRelay
```

- 8 Preview DSC.** Declares desired state instead of steps.

```
Configuration GfmBaseline { Import-DscResource -ModuleName
PSDesiredStateConfiguration
    Node 'localhost' { WindowsFeature NoSMB1 { Name='FS-SMB1'; Ensure='Absent' }
                        Service Spooler { Name='Spooler'; State='Stopped';
StartupType='Disabled' } } }
GfmBaseline -OutputPath C:\DSC; Test-DscConfiguration -Path C:\DSC -Verbose
```

- 9 Commit to Git.**

```
winget install Git.Git
Set-Location C:\Scripts; git init; git add *.ps1; git commit -m "Lab 28: weekly AD
report"
```

EXPECTED OUTPUT

```
LastTaskResult : 0
smtp4dev inbox: "AD weekly health 9/28/2026" with four HTML tables
Test-DscConfiguration -> InDesiredState : True (after Start-DscConfiguration)
```

TROUBLESHOOTING

Task result 0x1

Read the transcript in C:\Scripts\Logs. Commonly the gMSA lacks batch logon right or cannot load the AD module.

SmtpClient: unable to connect	smtp4dev not listening on 0.0.0.0:25 or firewall blocks it. New-NetFirewallRule -DisplayName SMTP -LocalPort 25 -Protocol TCP -Action Allow.
Get-ADReplicationPartnerMetadata errors	Run from a host with RSAT and network access to all DCs; -Target * queries every DC.

KNOWLEDGE CHECK

- Q1 Why is -WhatIf support important for production scripts?
- Q2 How does running as a gMSA eliminate a stored secret?
- Q3 What is the difference between imperative scripts and DSC?

SNAPSHOT ▸ L28-END-automation (DC01, DC03, SRV01)

LAB 29

LESSON 29

Monitoring, Health, and Performance

Create a Data Collector Set and a repeatable health-check script for the domain.

OBJECTIVES	• Read healthy dcdiag and repadmin output • Build a Data Collector Set for AD performance counters • Forward DC events to a collector • Automate a daily health check
PREREQUISITES	L28-END-automation. DC01, DC03 (and DC02 if rebuilt), SRV01 as event collector.

TASKS

1 Baseline healthy output. [DC01]

```
dcdiag /e /c /q
repadmin /replsummary
repadmin /showrepl * /csv | ConvertFrom-Csv | Where-Object 'Number of Failures' -
gt 0
repadmin /queue
```

2 Check time hierarchy. The PDC (now DC03) must sync from an external source; others show the PDC or another DC.

```
netdom query pdc
w32tm /monitor /domain:greenfieldcorp.lab
Invoke-Command DC03 { w32tm /query /source }
```

3 Move the reliable time source with the PDC. [DC03]

```
w32tm /config /manualpeerlist:"time.windows.com,0x8 pool.ntp.org,0x8"
/syncfromflags:manual /reliable:yes /update
Restart-Service w32time
Invoke-Command DC01 { w32tm /config /syncfromflags:domhier /reliable:no /update;
Restart-Service w32time }
```

4 Create a Data Collector Set. [DC01]

```
$counters = @(
  '\NTDS\LDAP Searches/sec', '\NTDS\LDAP Client Sessions', '\NTDS\DRS Inbound Bytes
Total/sec',
  '\NTDS\DS Directory Reads/sec', '\Security System-Wide Statistics\Kerberos
Authentications',
  '\Security System-Wide Statistics\NTLM Authentications', '\Processor(_Total)\%
Processor Time',
  '\Memory\Available MBytes', '\LogicalDisk(C:)\Avg. Disk sec/Read', '\DNS\Total
Query Received/sec')
$counters | Set-Content C:\PerfLogs\ad-counters.txt
logman create counter GFM-AD-Baseline -cf C:\PerfLogs\ad-counters.txt -si 00:00:15
-f bincirc -max 256 -o C:\PerfLogs\GFM-AD-Baseline
logman start GFM-AD-Baseline
logman query GFM-AD-Baseline
```

5 Generate load and review. After 10 minutes, stop and open the .blg in Performance Monitor.

```
1..500 | ForEach-Object { Get-ADUser -Filter * | Out-Null }
logman stop GFM-AD-Baseline
Get-ChildItem C:\PerfLogs\GFM-AD-Baseline*
```

6 Configure event forwarding. [SRV01 collector]

```
wecutil qc /q
# DC01 and DC03: add collector's computer account to Event Log Readers
Add-ADGroupMember 'Event Log Readers' -Members 'SRV01$'
```

7 Create a source-initiated subscription. Event Viewer > Subscriptions > Create: GFM-DC-Security, Source computer initiated, Computer group = Domain Controllers, events 4625, 4740, 4728, 4732, 4756, 5136, 5137, 4769 from Security; Error/Critical from Directory Service, DNS Server, DFS Replication. GPO DC-C-WEF-v1: Configure target Subscription Manager = Server=http://SRV01.greenfieldcorp.lab:5985/wsman/SubscriptionManager/WEC,Refresh=60.

```
wecutil gr GFM-DC-Security
Get-WinEvent -LogName ForwardedEvents -MaxEvents 10
```

8 Write C:\Scripts\Invoke-DailyADCheck.ps1.

```
$r = [ordered]@{}
$r.DcDiag = (dcdiag /e /q) -join [Environment]::NewLine
$r.ReplFail = (Get-ADReplicationFailure -Target * -Scope Domain -ErrorAction SilentlyContinue | Measure-Object).Count
$r.Sysvol = (Get-ADDomainController -Filter *).HostName | ForEach-Object { [pscustomobject]@{DC=$_;Ok=(Test-Path "\\$_\SYSVOL")} }
$r.TimeSkew = (w32tm /monitor /domain:greenfieldcorp.lab /nowarn) -match 'NTP:'
$r.Services = (Get-ADDomainController -Filter *).Name | ForEach-Object { Invoke-Command $_ { Get-Service NTDS,DNS,KDC,Netlogon,DFSR,W32Time | Where-Object Status -ne Running | Select-Object @{n='DC';e={$env:COMPUTERNAME}},Name } }
$r.FreeDisk = (Get-ADDomainController -Filter *).Name | ForEach-Object { Invoke-Command $_ { [pscustomobject]@{DC=$env:COMPUTERNAME;FreeGB=[int]((Get-PSDrive C).Free/1GB)} } }
$status = if (-not $r.DcDiag -and $r.ReplFail -eq 0 -and -not $r.Services) {'HEALTHY'} else {'ATTENTION'}
"$($Get-Date -Format s) $status ReplFailures=$(($r.ReplFail))" | Tee-Object -Append C:\Scripts\Logs\daily.log
$r | ConvertTo-Json -Depth 4 | Set-Content "C:\Scripts\Logs\daily-$(Get-Date -Format yyyyMMdd).json"
```

9 Schedule daily at 07:00.

```
Register-ScheduledTask GFM-DailyADCheck -Action (New-ScheduledTaskAction -Execute powershell.exe -Argument '-NoProfile -File C:\Scripts\Invoke-DailyADCheck.ps1') -Trigger (New-ScheduledTaskTrigger -Daily -At 7am) -User SYSTEM -RunLevel Highest
```

EXPECTED OUTPUT

```
Replication Summary Start Time: ...
Source DSA largest delta fails/total % error
DC01 05m:12s 0 / 5 0
DC03 04m:48s 0 / 5 0
[2026-09-23T07:00:02] HEALTHY ReplFailures=0
Name: GFM-AD-Baseline Status: Running
```

TROUBLESHOOTING

logman: counter path not found	NTDS counters exist only on DCs. Run logman on a DC.
ForwardedEvents empty	Collector computer account not in Event Log Readers on the DC, or WinRM URL wrong in the GPO. Check Microsoft-Windows-Eventlog-ForwardingPlugin/Operational on the DC.

w32tm /monitor shows large offset on one DC

It is syncing from VirtualBox Guest Additions time sync.

Disable: VBoxManage setextradata <vm>

"VBoxInternal/Devices/VMMDev/0/Config/GetHostTimeDisabled"
1.

KNOWLEDGE CHECK

- Q1 What does a healthy repadmin /replsummary look like?
- Q2 Why must the PDC Emulator of the forest root be the only DC configured with an external time source?
- Q3 Which three counters would you alert on first for a DC, and why?

SNAPSHOT ▸ L29-END-monitoring (DC01, DC03, SRV01)

LAB 30

LESSON 30

Hybrid Identity with Entra ID

Connect the lab domain to an Entra ID tenant and configure hash synchronization.

OBJECTIVES

- Prepare UPN suffixes and filter sync scope
- Install Entra Connect Sync with password hash synchronization
- Verify synchronized users and password sync
- Check sync health and understand source of authority

PREREQUISITES

L29-END-monitoring. A free Entra ID test tenant (Microsoft 365 Developer Program or trial) with a verified custom domain, e.g. greenfieldlab.example. Never connect the lab to a production tenant. SRV01 has internet via GFM-NAT.

TASKS

- 1 **Add a routable UPN suffix. [DC01]** .lab is not routable; users need a UPN matching the verified tenant domain.

```
Set-ADForest greenfieldcorp.lab -UPNSuffixes @{Add='greenfieldlab.example'}
Get-ADUser -SearchBase
'OU=Engineering,OU=Users,OU=Greenfield,DC=greenfieldcorp,DC=lab' -Filter * |
ForEach-Object {
    Set-ADUser $_ -UserPrincipalName "$($_.SamAccountName)@greenfieldlab.example" -
    EmailAddress "$($_.SamAccountName)@greenfieldlab.example" }
```

- 2 **Check for sync-blocking attribute issues.** Duplicate UPNs/proxyAddresses and invalid characters cause export errors.

```
Get-ADUser -Filter * -Properties proxyAddresses, mail | Group-Object
UserPrincipalName | Where-Object Count -gt 1
Get-ADUser -Filter * | Where-Object UserPrincipalName -match '^[a-zA-Z0-9.@_-]'
```

- 3 **Create a sync-scope OU filter plan.** Sync OU=Users only. Do not sync OU=Admin (Tier 0 identities stay on-premises only).

- 4 **Enable TLS 1.2 for .NET on SRV01. [SRV01]**

```
$k='HKLM:\SOFTWARE\Microsoft\.NETFramework\v4.0.30319'
New-ItemProperty $k -Name SchUseStrongCrypto -Value 1 -PropertyType DWord -Force
New-ItemProperty $k -Name SystemDefaultTlsVersions -Value 1 -PropertyType DWord -Force
```

- 5 **Install Entra Connect Sync.** Download from the Entra admin center > Entra Connect. Choose Customize: Password Hash Synchronization; Seamless SSO off for now; sign in with a tenant Hybrid Identity Administrator; add forest greenfieldcorp.lab with "Create new AD account"; UPN suffix greenfieldlab.example; Domain and OU filtering = OU=Users only; keep "Start the synchronization process" checked.

- 6 **Verify the connector and scheduler. [SRV01]**

```
Import-Module ADSync
Get-ADSyncScheduler
Get-ADSyncConnector | Select-Object Name, Type
Start-ADSyncSyncCycle -PolicyType Delta
```

- 7 **Verify users in the cloud.** Entra admin center > Users: Engineering users show "On-premises sync enabled: Yes".

```
Install-Module Microsoft.Graph.Users -Scope CurrentUser
Connect-MgGraph -Scopes 'User.Read.All'
Get-MgUser -Filter "onPremisesSyncEnabled eq true" -ConsistencyLevel eventual -
CountVariable n -All | Select-Object DisplayName, UserPrincipalName
$n
```

- 8 **Test password hash sync.** Change maria.lopez password on-premises, wait ~2 minutes, sign in at myaccount.microsoft.com with the new password.

```
Set-ADAccountPassword maria.lopez -Reset -NewPassword (Read-Host -AsSecureString)
Get-WinEvent -LogName Application -MaxEvents 30 | Where-Object { $_.ProviderName -
eq 'Directory Synchronization' -and $_.Id -in 656,657 }
```

- 9 **Test source of authority.** Try editing a synced user job title in the Entra portal: it is read-only because AD is authoritative.

10 Review Connect Health and Conditional Access. Entra admin center > Entra Connect > Connect Sync: last sync time and errors. Create (report-only) a Conditional Access policy requiring MFA for Engineering users; compare its scope to on-premises GPO.

EXPECTED OUTPUT

```
AllowedSyncCycleInterval : 00:30:00
SyncCycleEnabled          : True
Name                      Type
greenfieldcorp.lab        AD
<tenant>.onmicrosoft.com - AAD Extensible2
Event 657: Password change result: Success
```

TROUBLESHOOTING

Users sync as user@<tenant>.onmicrosoft.com	UPN suffix not verified in the tenant, or users still have @greenfieldcorp.lab. Verify the domain and fix UPNs, then run a delta sync.
Wizard fails: unable to connect to Entra ID	TLS 1.2 registry values missing, or proxy/DNS forwarder issue. Test with Invoke-WebRequest https://login.microsoftonline.com.
Password changes do not sync	The AD connector account lacks Replicate Directory Changes (All). Re-run the wizard or grant with Set-ADSyncPasswordHashSyncPermissions.

KNOWLEDGE CHECK

Q1 Compare PHS, PTA, and federation in terms of on-premises dependency.

Q2 Why keep Tier 0 accounts out of synchronization?

Q3 Which hybrid identity decisions are hardest to reverse later?

SNAPSHOT ▶ L30-END-hybrid (DC01, DC03, SRV01)

Troubleshooting Methodology

Work through a set of pre-broken lab scenarios and diagnose each to root cause.

OBJECTIVES

- Apply a structured sequence: DNS, time, replication, permissions
- Diagnose five injected faults to root cause
- Record findings in a repeatable checklist

PREREQUISITES

L30-END-hybrid. Take L31-START on all VMs. Ideally a partner runs the break script so you do not know which scenario is which; otherwise run scenarios in random order.

TASKS

- 1 Use the diagnostic sequence for every scenario.** 1 Define the symptom and scope. 2 Is it the network? (Test-NetConnection). 3 DNS (Resolve-DnsName, SRV records). 4 Time (w32tm). 5 Replication (repadmin). 6 Authentication/permissions (klist, events). 7 Fix, verify, document.
- 2 Save the break script (partner only). [host]** Each scenario lists the command to run and on which VM.

```
# S1 [CL01] DNS: point client at a public resolver
Set-DnsClientServerAddress -InterfaceAlias Ethernet -ServerAddresses 1.1.1.1
# S2 [DC03] Time: skew the clock 20 minutes and stop time service
Stop-Service w32time; Set-Date (Get-Date).AddMinutes(20)
VBoxManage setextradata DC03
"VBoxInternal/Devices/VMMDev/0/Config/GetHostTimeDisabled" 1
# S3 [DC01] Replication: block RPC to DC03
New-NetFirewallRule -DisplayName 'LAB-S3' -Direction Outbound -RemoteAddress
10.10.10.12 -Action Block
# S4 [DC01] GPO: remove Read for Authenticated Users on the lockdown GPO
Set-GPPermission -Name USR-U-DesktopLockdown-v1 -TargetName 'Authenticated Users'
-TargetType Group -PermissionLevel None -Replace
# S5 [DC01] Secure channel: reset SRV01's computer account password in AD only
Get-ADComputer SRV01 | Set-ADAccountPassword -Reset -NewPassword (ConvertTo-
SecureString 'Br0ken.Trust!2025' -AsPlainText -Force)
```

- 3 Scenario A: "Users on CL01 cannot log on — domain not available."** Check DNS first.

```
ipconfig /all | Select-String 'DNS Servers'
nltest /dsgetdc:greenfieldcorp.lab
Resolve-DnsName -Type SRV _ldap._tcp.dc._msdcs.greenfieldcorp.lab
```

- 4 Scenario B: "Kerberos errors on DC03, replication failing."**

```
w32tm /stripchart /computer:DC01 /samples:3 /dataonly
Get-WinEvent -FilterHashtable @{LogName='System';ProviderName='Microsoft-Windows-Security-Kerberos'} -MaxEvents 5
repadmin /showrepl DC03
```

- 5 **Scenario C: "Changes on DC01 do not appear on DC03."** Read the error code in repadmin: 1722 = RPC unavailable (network/firewall), 8453 = access denied, 8606 = lingering objects, 1256 = remote system unavailable.

```
repadmin /replsummary
repadmin /showrepl DC01 /errorsonly
Test-NetConnection 10.10.10.12 -Port 135
Get-NetFirewallRule | Where-Object DisplayName -like 'LAB-*
```

- 6 **Scenario D: "Desktop lockdown stopped applying."**

```
gpresult /r /scope user
Get-WinEvent -LogName 'Microsoft-Windows-GroupPolicy/Operational' -MaxEvents 30 |
Where-Object Level -le 3
Get-GPPermission -Name USR-U-DesktopLockdown-v1 -All | Select-Object Trustee,
Permission
```

- 7 **Scenario E: "SRV01: trust relationship between this workstation and the primary domain failed."**

```
Test-ComputerSecureChannel -Verbose
nltest /sc_query:greenfieldcorp.lab
Test-ComputerSecureChannel -Repair -Credential (Get-Credential
GREENFIELD\Administrator)
```

- 8 **Separate AD, DNS, and network.** If Test-NetConnection to the DC IP fails, it is network. If IP works but name fails, it is DNS. If both work but logon fails, it is AD/authentication.

```
Test-NetConnection 10.10.10.10 -Port 389
Test-NetConnection DC01.greenfieldcorp.lab -Port 88
```

- 9 **Record each diagnosis.** For every scenario write: symptom, first failing check, root cause, fix, verification command, time to resolve. Save as C:\Scripts\runbooks\troubleshooting-log.md.
- 10 **Revert.** Restore L31-START snapshots, or reverse each change manually and confirm dcdiag /e /q is clean.

EXPECTED OUTPUT

```
A: DNS Servers: 1.1.1.1 -> nltest: ERROR_NO_SUCH_DOMAIN -> fix DNS to 10.10.10.10
B: stripchart offset +1200s -> Kerberos KRB_AP_ERR_SKEW
C: repadmin error 1722 (0x6ba) The RPC server is unavailable
D: gpresult: USR-U-DesktopLockdown-v1 "Filtering: Inaccessible"
E: Test-ComputerSecureChannel False -> Repair -> True
```

TROUBLESHOOTING

Fixing symptoms before finding the cause	Stop rebooting. Follow the sequence and write down the first check that fails.
Scenario C fixed but replication still behind	Force it: repadmin /syncall /AdeP, then repadmin /replsummary.
Clock skew recurs after fix	Re-enable w32time and check VirtualBox host time sync settings.

KNOWLEDGE CHECK

- Q1 Why check DNS and time before replication and permissions?
- Q2 What does repadmin error 8453 tell you, and what would you check next?
- Q3 When should you escalate instead of continuing to troubleshoot?

SNAPSHOT ▸ L31-END-troubleshooting (DC01, DC03, SRV01, CL01)

LAB 32

LESSON 32

Design, Best Practices, and Career Direction

Design a complete directory for a new fictional organization and document it end to end.

OBJECTIVES	• Produce a greenfield AD design from requirements • Write a design document a successor can use • Create daily, weekly, and monthly runbooks • Consolidate a 2025 security baseline checklist
PREREQUISITES	All prior labs complete. Keep the lab as a permanent practice environment: export VMs with VBoxManage export after this lab.

TASKS

- 1 Read the brief.** Harbor Freight Cooperative (fictional): 1,200 users; HQ in Toledo, 3 warehouses on 50 Mbps links, 1 warehouse with no IT staff; 60 servers; Microsoft 365; a 2012 R2 forest being retired; PCI scope for payment systems.

- 2 **Design the forest and domain model.** Decide single forest/single domain vs. separate forest for PCI. Justify in one paragraph referencing the forest as the security boundary.
- 3 **Design OUs, naming, and groups.** Reuse the Appendix C conventions. Include Tier 0/1/2, a PCI OU if in-forest, and the AGDLP resource model.
- 4 **Design sites and DCs.** Sites, subnets, site links with costs; DC count per site; RODC for the unstaffed warehouse; FSMO and GC placement; DNS forwarders.
- 5 **Design security.** Map each Part IV control: baseline GPOs, LDAP signing/channel binding, SMB signing, LAPS, Protected Users, dMSA creation restrictions, auditing to SIEM, PKI hierarchy.
- 6 **Design hybrid identity and recovery.** PHS vs PTA choice, sync scope, Tier 0 exclusion; backup schedule, Recycle Bin, forest recovery owner; functional level 2025 and 32k pages timeline.
- 7 **Generate current-state documentation from the lab as an example.** Use this as a template appendix in your design doc.

```
$out='C:\Scripts\Docs'; New-Item $out -ItemType Directory -Force | Out-Null
Get-ADForest | Select-Object Name, ForestMode, SchemaMaster, DomainNamingMaster,
Sites, UPNSuffixes | Out-File "$out\forest.txt"
Get-ADDomain | Select-Object DNSRoot, DomainMode, PDCEmulator, RIDMaster,
InfrastructureMaster | Out-File "$out\domain.txt"
Get-ADDomainController -Filter * | Select-Object Name, IPv4Address, Site,
IsGlobalCatalog, OperatingSystem | Export-Csv "$out\dcs.csv" -NoTypeInfoation
Get-ADReplicationSite -Filter * | Select-Object Name | Export-Csv "$out\sites.csv"
-NoTypeInfoation
Get-ADReplicationSubnet -Filter * | Select-Object Name, Site | Export-Csv
"$out\subnets.csv" -NoTypeInfoation
Get-ADOrganizationalUnit -Filter * | Select-Object DistinguishedName | Export-Csv
"$out\ous.csv" -NoTypeInfoation
Get-GPO -All | Select-Object DisplayName, GpoStatus, ModificationTime | Export-Csv
"$out\gpos.csv" -NoTypeInfoation
Get-GPOReport -All -ReportType Html -Path "$out\gpos.html"
Get-ADFineGrainedPasswordPolicy -Filter * | Select-Object Name, Precedence,
MinPasswordLength | Export-Csv "$out\psos.csv" -NoTypeInfoation
```

- 8 **Write the runbooks.** Daily: Invoke-DailyADCheck review, failed logons/lockouts, backup success. Weekly: AD report email, privileged group review, patch status. Monthly: stale accounts cleanup, GPO change review, restore test of one object, CRL/cert expiry check, DSRM password verification.
- 9 **Complete the 2025 baseline checklist.** Confirm each item in your lab and in the design: SMB signing required; LDAP signing + channel binding; LDAPS with TLS 1.3 where

supported; LAPS with encryption; no unconstrained delegation except DCs; Protected Users for Tier 0; dMSA creation restricted and audited (5137/5136 on msDS-ManagedAccountPrecededByLink); RC4 disabled; Recycle Bin on; 32k pages on fresh 2025 DCs; baseline GPOs linked; Schema Admins empty.

```
Get-SmbServerConfiguration | Select-Object RequireSecuritySignature
Get-ItemProperty HKLM:\SYSTEM\CurrentControlSet\Services\NTDS\Parameters | Select-Object LDAPServerIntegrity, LdapEnforceChannelBinding
Get-ADComputer -Filter 'TrustedForDelegation -eq $true' | Select-Object Name
Get-ADGroupMember 'Schema Admins'
Get-ADOptionalFeature -Filter * | Select-Object Name, EnabledScopes
```

10 Export the lab for permanent practice. [host]

```
'DC01','DC03','SRV01','SRV02','CL01' | ForEach-Object { VBoxManage controlvm $_ acpipowerbutton }
VBoxManage export DC01 DC03 SRV01 SRV02 CL01 -o C:\LabExport\GFM-Lab-Complete.ova --manifest
```

11 Write your 90-day plan. Days 1–30: inventory and health baseline of your real environment. 31–60: close top three security gaps from the checklist. 61–90: automate reporting and document recovery. Map skills to certification objectives you are targeting.

EXPECTED OUTPUT

```
C:\Scripts\Docs\ contains forest.txt, domain.txt, dcs.csv, sites.csv, subnets.csv, ous.csv, gpos.csv, gpos.html, psos.csv
Design document: 8-12 pages covering forest, OU, sites, security, hybrid, recovery, runbooks
GFM-Lab-Complete.ova exported with manifest
```

TROUBLESHOOTING

Design mirrors Greenfield exactly	Re-read the brief: bandwidth, the unstaffed site, and PCI change the answers. Justify each difference.
VBoxManage export fails: VM is running	Wait for all VMs to show powered off (VBoxManage list runningvms is empty).

KNOWLEDGE CHECK

Q1 What would push you to a second forest rather than a second domain?

Q2 Which three documents would a successor need first?

Q3 Why does on-premises AD still matter in a cloud-first organization?

SNAPSHOT ▸ L32-END-course-complete (all)

E

Command cheat sheet

SERVER SETUP & REMOTING

<code>sconfig</code>	Server Core configuration menu
<code>New-NetIPAddress -InterfaceAlias LAN -IPAddress 10.10.10.10 -PrefixLength 24 -DefaultGateway 10.10.10.1</code>	Static IPv4
<code>Set-DnsClientServerAddress -InterfaceAlias LAN -ServerAddresses 10.10.10.10,10.10.10.11</code>	DNS client servers
<code>Rename-Computer -NewName SRV01 -Restart</code>	Rename host
<code>Add-Computer -DomainName greenfieldcorp.lab -Restart</code>	Join domain
<code>Enable-PSRemoting -Force</code>	Enable WinRM
<code>Invoke-Command -ComputerName DC01,DC03 -ScriptBlock { ... }</code>	One-to-many remoting
<code>Install-WindowsFeature <Name> -IncludeManagementTools</code>	Add role/feature

DOMAIN CONTROLLERS

<code>Install-ADDSForest -DomainName greenfieldcorp.lab -InstallDns</code>	New forest
<code>Install-ADDSDomainController -DomainName greenfieldcorp.lab -InstallDns</code>	Additional DC
<code>Uninstall-ADDSDomainController -DemoteOperationMasterRole</code>	Demote DC
<code>dcdiag /e /c /q</code>	Full health test, errors only
<code>repadmin /replsummary</code>	Replication summary
<code>repadmin /showrepl <DC> /errorsonly</code>	Per-DC replication errors
<code>repadmin /syncall /AdeP</code>	Force replication, all partitions
<code>nltest /dsgetdc:greenfieldcorp.lab</code>	DC locator test
<code>nltest /dsgetsites</code>	Current site

FSMO & TIME

<code>netdom query fsmo</code>	List role holders
<code>Move-ADDirectoryServerOperationMasterRole -Identity DC03 -OperationMasterRole 0,1,2,3,4</code>	Transfer roles (add -Force to seize)
<code>w32tm /query /source</code>	Time source
<code>w32tm /monitor /domain:greenfieldcorp.lab</code>	Offsets across DCs
<code>w32tm /resync /rediscover</code>	Force resync

DNS

<code>Resolve-DnsName -Type SRV _ldap._tcp.dc._msdcs.greenfieldcorp.lab</code>	Locate DCs
--	------------

<code>Get-DnsServerZone</code>	List zones
<code>Add-DnsServerPrimaryZone -NetworkId '10.10.10.0/24' -ReplicationScope Forest</code>	Reverse zone
<code>Set-DnsServerForwarder -IPAddress 1.1.1.1,9.9.9.9</code>	Forwarders
<code>ipconfig /registerdns ; Restart-Service Netlogon</code>	Re-register A / SRV records
USERS, GROUPS, COMPUTERS	
<code>New-ADUser -Name ... -SamAccountName ... -UserPrincipalName ... -Path <OU></code>	Create user
<code>Import-Csv users.csv ForEach-Object { New-ADUser ... }</code>	Bulk create
<code>New-ADGroup <Name> -GroupScope Global DomainLocal Universal</code>	Create group
<code>Get-ADGroupMember <Group> -Recursive</code>	Effective membership
<code>Search-ADAccount -AccountInactive -TimeSpan 90.00:00:00</code>	Stale accounts
<code>Search-ADAccount -LockedOut Unlock-ADAccount</code>	Unlock all locked accounts
<code>Test-ComputerSecureChannel -Repair</code>	Repair trust
GROUP POLICY	
<code>gpupdate /force</code>	Refresh policy
<code>gpresult /r /scope user computer</code>	Applied GPOs summary
<code>gpresult /h report.html</code>	Full RSoP report
<code>New-GPO <Name> New-GPLink -Target <OU></code>	Create and link
<code>Set-GPLink -Name <GPO> -Target <OU> -Enforced Yes</code>	Enforce link
<code>Get-GPInheritance -Target <OU></code>	Precedence at an OU
<code>Backup-GPO -All -Path C:\GPOBackup</code>	Back up all GPOs
KERBEROS, ACCOUNTS, SECURITY	
<code>klist ; klist purge ; klist -li 0x3e7 purge</code>	View / purge tickets (user, computer)
<code>setspn -S <SPN> <account> ; setspn -X</code>	Add SPN safely; find duplicates
<code>Add-KdsRootKey -EffectiveImmediately</code>	KDS key for gMSA/dMSA (wait 10h in prod)
<code>New-ADServiceAccount <n> -CreateDelegatedServiceAccount</code>	Create dMSA
<code>Start-/Complete-ADServiceAccountMigration</code>	Migrate legacy account to dMSA
<code>Get-LapsADPassword -Identity <PC> -AsPlainText</code>	Retrieve LAPS password
<code>auditpol /get /category:*</code>	Effective audit policy
RECOVERY & LIFECYCLE	
<code>wbadmin start systemstatebackup -backupTarget:E:</code>	System state backup

Get-ADObject -Filter 'isDeleted -eq \$true' - IncludeDeletedObjects Restore-ADObject	Recycle Bin restore
bcdedit /set safeboot dsrepair	Boot to DSRM
ntdsutil "activate instance ntds" "authoritative restore" "restore subtree <DN>"	Authoritative restore
Set-ADForestMode -ForestMode Win2025Forest	Raise forest level (one-way)
Enable-ADOptionalFeature 'Database 32k Pages Feature' -Scope ForestOrConfigurationSet -Target <forest>	Enable 32k pages (irreversible)

VIRTUALBOX (HOST)

VBoxManage natnetwork add --netname GFM-NAT --network 10.10.10.0/24 --dhcp off	Create lab network
VBoxManage clonevm TPL-WS2025 --snapshot <snap> -- options link --name <VM> --register	Linked clone
VBoxManage snapshot <VM> take L##-END-<state>	Take snapshot
VBoxManage snapshot <VM> restore <name>	Restore snapshot (VM off)
VBoxManage modifyvm <VM> --nested-hw-virt on	Nested virtualization